

PERFORMANCE INVESTIGATIONS ON PARABOLIC TROUGH SOLAR THERMAL COLLECTOR WITH SECONDARY REFLECTOR

PROJECT REPORT

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to

*the APJ Abdul Kalam Technological University
in partial fulfillment of the requirements for the award of Bachelor of
Technology In Mechanical Stream - Production Engineering .*



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DEPARTMENT OF MECHANICAL ENGINEERING

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CERTIFICATE

Certified that this report entitled “*PERFORMANCE INVESTIGATIONS ON PARABOLIC TROUGH SOLAR THERMAL COLLECTOR WITH SECONDARY REFLECTOR*” is the report of the project presented by **ABRAHAM KORAH JOHN (TKM18MP007)**, **DEVU NANDAN (TKM18MP029)**, **JOBSON K JAMES (TKM18MP031)**, **NIRAJ N NAIR (TKM18MP043)** during 2018-2022 in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Mechanical Stream – Production Engineering of the APJ Abdul Kalam Technological University.

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DECLARATION

We hereby declare that this project report entitled “Performance Investigations on Parabolic Trough Solar Thermal Collector With Secondary Reflector” is our bonafide work carried out under the supervision of Dr. Baiju V, Assistant Professor, Department of Mechanical Engineering, T K M College of Engineering, Kollam. We declare that to the best of our knowledge the work reported herein does not form part of any other project report or dissertation based on which a degree or award was conferred on an earlier occasion to any other candidate. The content of this report is not being presented by any other student of this or any other University for the award of a degree.

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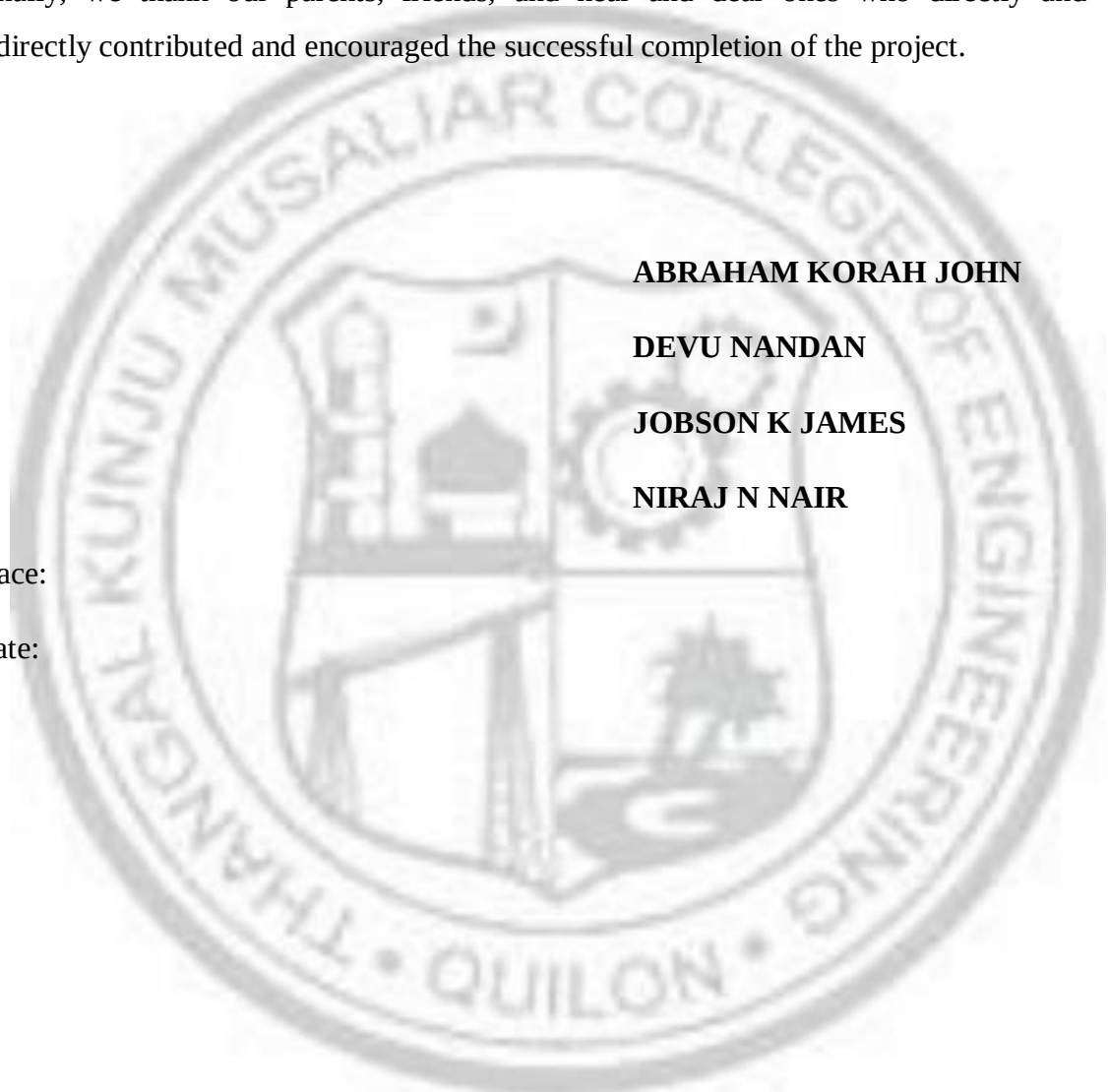
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ABSTRACT

Concentrating solar power (CSP) systems with parabolic trough collectors have been proven to be successful and are widely used commercially. The receiver tube, the heat collecting element, of a solar parabolic trough collector, is subjected to large circumferential temperature gradients due to the non-uniform concentration of solar flux and low thermal conductivity of conventional working fluids. The thermal stresses caused by thermal gradients can bend the absorber tube and eventually lead to structural failure. Therefore, analysis of the receiver tube for identifying the uneven heat flux distribution and methods to mitigate the thermal gradients are very essential. In the present study, ray-tracing analyses are carried out to evaluate the heat flux distribution on the receiver surface and output power of the PTC using Tonatiuh software. Optical simulations have been conducted for secondary reflectors with two strips to infinite strips and compared the results to optimize the secondary reflector for a combination of better power output with desirable uniformity of heat flux distribution. It is found that the uniformity and power output increase as the number of strips increases and beyond a certain level the uniformity and power put decline. For all secondary reflectors evaluated for a PTC with a focal length of 1m, the maximum uniformities are achieved when the secondary reflector positions are in between 0.996m and 1.012m. The selected shape uniformity was obtained for PTC with secondary reflector is 77.066% greater as compared to the PTC without a secondary reflector. The maximum increase in power output for the PTC with secondary reflector is 43.58% than the conventional PTC. Also done the prediction modelling of the PTC with the selected shape of secondary reflector using Artificial Neural Network and found out an accuracy of 60%

KEYWORDS: Parabolic trough solar collector (PTC); Monte Carlo Ray Tracing (MCRT); Optical simulation; Tonatiuh; Artificial Neural Network (ANN); Support Vector Machine (SVM)

CONTENTS

Title	Page Number
List of Figures	vi
List of Tables	vii
Chapter-1. Introduction	
1.1 General Background	1
1.2 The Sun and The Earth	1
1.2.1 Solar Spectrum and Solar Radiation	2
1.2.2 Direct and Diffuse Solar Radiation	3
1.3 Sun and Earth Angles	3
1.4 Solar Energy in India	4
1.5 Applications of Solar Energy	5
1.6 Solar Energy Collectors	5
1.6.1 Types of Solar Thermal Collectors	5
1.6.1.1 Flat-plate Collectors	6
1.6.1.2 Compound Parabolic Collectors	7
1.6.1.3 Evacuated Tube Collectors	8
1.6.1.4 Parabolic Trough Collectors	8
1.6.1.5 Linear Fresnel Reflector	9
1.6.1.6 Parabolic Dish Collector	10
1.6.1.7 Heliostat Field Collector	10
1.7 Sun Tracking System	11
1.8 Heat Collection Element (HCE)	13
1.9 Heat Transfer Fluids of PTSC	13
1.10 Storage of Thermal Energy	16
1.11 Objectives	16

1.12 Scope of the study	17
1.14 Report Outline	17
Chapter–2 Literature Review	18
2.1 Structural design and modification of PTC	19
2.2 Geometry modification of receiver tubes	20
2.3 Advanced heat transfer fluids	21
Chapter–3 Methodology	23
Chapter-4 Design of a Parabolic Trough Collector	24
4.1 Basic Components of PTC	24
4.1.1 Collector	24
4.1.2 Absorber Tube	24
Chapter–5. Optical Simulation	28
5.1 Introduction	28
5.2 Secondary reflector	28
5.3 Monte Carlo Ray Tracing (MCRT) technique	29
5.4 Ray tracing tools	30
5.5 Parameters of PTC	31
5.6 Optical model of PTC	32
5.7 Results and discussion	33
5.7.1 Ray Tracing simulation	33
5.7.2 Heat flux	34
Chapter-6 Design Optimization of Secondary Reflector	39
6.1 Introduction	39
6.2 Shape Selection	39

Chapter-7 Experimentation	48
7.1 Weather Station	48
Chapter-8 Machine Learning in PTC	49
8.1 Need for Machine Learning	49
8.2 Artificial Neural Networks	49
8.3 Parameter Settings	49
8.4 Feature Study	50
8.4.1 Heat Map of the correlations	50
8.4.2 Pairplot	51
8.5 Results and observations	51
8.6 Solutions	52
Chapter-9 Cost analyses	55
Chapter-10 Conclusion	56
10.1 Conclusion	56
10.2 Scope for future work	57
10.3 Summary	57
References	58

LIST OF FIGURES

Title	Page Number
Fig. 1.1 Spectral distribution of extra-terrestrial solar radiation	2
Fig. 1.2 Direct and diffuse radiations	3
Fig. 1.3 Sun and Earth angles	3
Fig. 1.4 Flat plate collector	7
Fig. 1.5 Compound parabolic collectors	7
Fig. 1.6 Evacuated tube collector	8
Fig. 1.7 Parabolic trough collector	9
Fig. 1.8 Linear Fresnel reflector	10
Fig. 1.9 Parabolic dish collector	10
Fig. 1.10 Heliostat field collector	11
Fig. 1.11 Schematic diagram of an active sun-tracking system	12
Fig. 1.12 Schematic of a receiver tube	13
Fig. 3.1 Conceptual framework of the study	23
Fig. 4.1 A typical parabolic trough collector	24
Fig. 4.2 Nomenclature of parabolic trough collectors	25
Fig. 5.1 The schematic diagram of the parabolic trough collector	29
Fig. 5.2 Schematic diagram of PTC with secondary reflector	31
Fig. 5.3 Traced paths of rays (a) without reflector, (b) with reflector	34
Fig. 5.4 Flux distribution on absorber surface (a) above focal axis, (b) at focal axis, (c) below focal axis	35
Fig. 5.5 Heat flux distribution for receiver above the focal axis	36
Fig. 5.6 Heat flux distribution for a receiver at the focal axis	37
Fig. 5.7 Heat flux distribution for the receiver below the focal axis	37
Fig 6.1 Flux distribution at periphery of receiver tube	42
Fig 6.2 Power output with different shapes	46
Fig. 6.3 Uniformity at different (a) Shapes, (b) Reflector positions	47
Fig 7.1 Conventional PTC at TKMCE	48

Fig 7.2a PTC with selected shape (6 strip 144 ⁰)	49
Fig 7.2b PTC with selected shape (6 strip 144 ⁰)	49
Fig. 8.1 Heat map of the parameters taken for the ANN of the PTC	51
Fig. 8.2 Pairplot of the parameters taken for the ANN of the PTC	52

LIST OF TABLES

Title	Page Number
Table 4. 1 Dimensions of PTC	27
Table 5.1 Parameters of PTC	32
Table 6.1 Optical simulations for different secondary reflector geometries	39
Table 6.2 Flux distribution, uniformity, and power output for different configurations	42
Table 6.3 Uniformity and output power for different secondary reflector configurations	46
Table 7.1 Experimental data (date 23/03/2022) for the secondary reflector with 6 strip (144^0)	48
Table 9.1 Cost analyses of the project	55

CHAPTER 1: INTRODUCTION

1.1 GENERAL BACKGROUND

Worldwide energy consumption has been growing daily due to the rise in the world's population and the increased industrial activities. Fossil fuels supply 78% of the energy demand in the world. It is widely acknowledged that the continued utilization of fossil fuels is not sustainable owing to the harm these fuels cause to the environment. Fossil fuels are non-renewable and the CO₂ emitted from power plants and energy systems using fossil fuels has exacerbated global warming and climate change (Rawani, Sharma and Singh, 2017). Moreover, there are increasing concerns related to the security of energy supply as well as the need to meet the ever-increasing demand for energy. To meet this increasing demand for energy with little or no harm to the environment while ensuring the security of energy supply, alternative, widely available, clean, and renewable energy sources have to be sought and developed. Some of the existing renewable energy sources include geothermal, wind energy, bioenergy, and solar energy. These resources provide the needed shift from conventional fossil-based resources.

Among the many alternative renewable energy sources, solar energy is the most important due to it being sustainable and environment-friendly energy source. The industrial age gave us an understanding of sunlight as an energy source. It is a clean and free source of energy. It can be harnessed even in rural and remote areas for electricity, heating, and refrigeration purposes.

1.2 THE SUN AND THE EARTH

The sun is the largest and the most massive object in the solar system. The diameter of the sun is about 696,000 kilometres (km) which is about 109 times that of earth. The earth is the third planet from the sun, and its diameter is around 12,756 kilometres. The distance between the sun and the earth varies slightly over the year and the average distance is about 151.65 million kilometres. The surface temperature of the sun is about 5800 K and closely approximates a blackbody (Okonkwo, Abid and Ratlamwala, 2018).

Solar energy is an infinite source of energy. The sun delivers approximately 1,575–49,837 exajoules (S. Ebhota and Jen, 2019) of energy on the earth which is far more energy than we need to power everything on earth. Solar energy is the largest untapped energy source, only 1.01% of this energy is currently utilized. Solar energy could supply all the present and future energy needs of the world continually. This makes solar energy one of the most promising renewable energy sources today. Therefore, the utilization of solar energy and its technologies have received much attention.

1.2.1 Solar spectrum and solar radiation

The energy in solar radiation comes in the form of electromagnetic waves of a wide spectrum, which is categorized further into ultraviolet, visible, and infrared spectrums. The spectral distribution of extraterrestrial solar radiation corresponding to the solar constant of 1367 W/m^2 is illustrated in Fig. 1.1 (S. Ebhota and Jen, 2019). It will be seen that the spectral value first increases sharply with wavelength and passes through a maximum at a wavelength of $0.48 \mu\text{m}$.

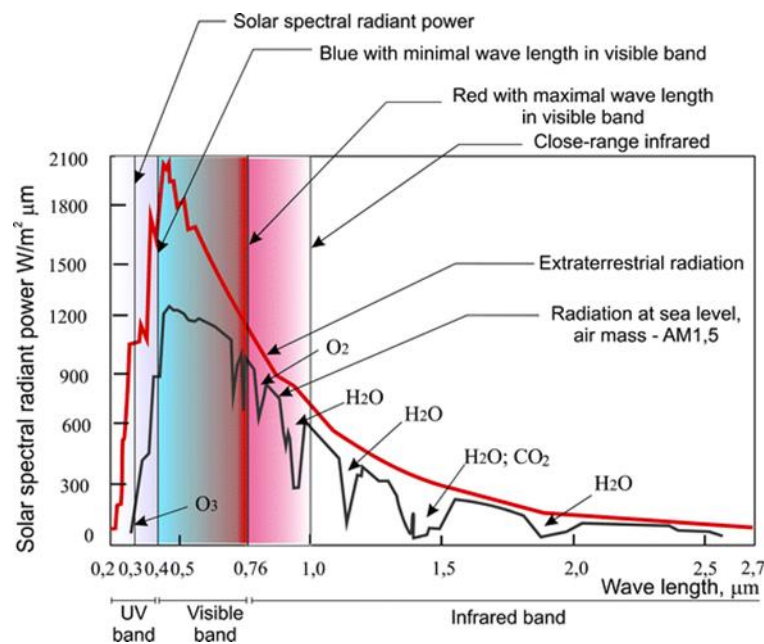


Fig. 1.1 Spectral distribution of extra-terrestrial solar radiation

About 48% of this radiation falls in the visible region, whereas about 45.6% is in the infrared region and the rest in the ultraviolet spectrum.

1.2.2 Direct and diffuse solar radiation

The solar radiation that fills the sky can be direct, diffuse, or reflected radiation as illustrated in Fig. 1.2. Direct radiation also called beam radiation is the solar radiation traveling on a straight line from the sun down to the surface of the earth. Diffuse radiation, on the other hand, means the sunlight that has been scattered by molecules and particles in the atmosphere but that has still made it down to the surface of the earth. Direct radiation has a definite direction, but diffuse radiation is randomly spread across a surface. This is because when the radiation is direct, the rays are traveling in the same direction but an object in front of it can block the rays and scatter them in different directions.

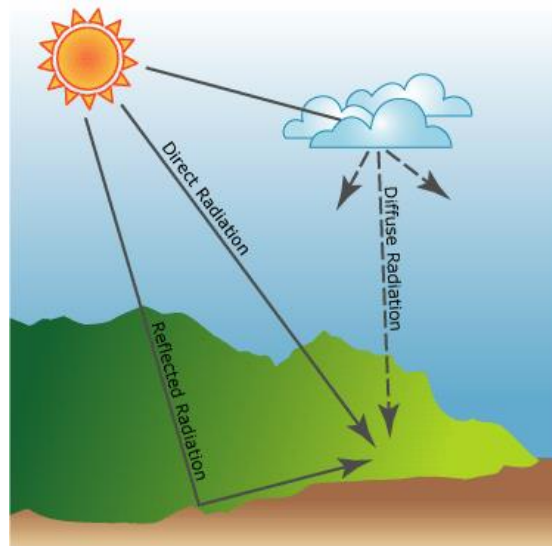


Fig. 1.2 Direct and diffuse radiations

1.3 SUN AND EARTH ANGLES

To track the sun throughout the day for every day of the year, there are several geometric relationships for the position of the collector that is needed to be known. These angles are used to define the sun's position in relation to about

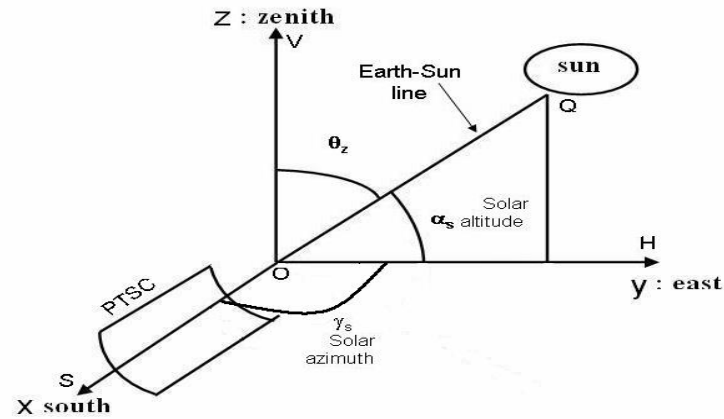


Fig. 1.3 Sun and Earth angles

Different Sun-Earth angles are:

1. Zenith angle (θ_z): It is the angle QOV between the sun's rays and a line perpendicular to the horizontal surface at O.
2. Solar altitude angle (α_s): It is the angle QOH on a vertical plane between the sun's rays and its projection on the horizontal plane i.e., the complement of the zenith angle. It follows that:

$$\alpha_s + \theta_z = 90^\circ \quad (1.1)$$

3. Solar azimuth angle (γ_z): It is the angle HOS, and it is the angular displacement from the south to the horizontal projection of the sun's rays.
4. Latitude: It is a geographic coordinate that specifies the north-south position of a point on the earth's surface.
5. Longitude: It is a geographic coordinate that specifies the west-east position of a point on the earth's surface.
6. Hour angle: The hour angle of a point is the angle between two planes: one containing the earth's axis and the meridian plane, and the other containing the earth's axis and the given point.
7. Sun's declination angle: It is the angular distance of a sun's ray north (or south) of the equator. It is the angle between the centre of the sun to the center of the earth and the projection of this line upon the earth's equatorial plane.
8. Angle of incidence: It is the angle between the sun's ray and a normal to the earth's surface.

1.4 SOLAR ENERGY IN INDIA

India is endowed with vast solar energy potential. About 5000 trillion kWh per year of energy is incident over India's land area with most parts receiving 4-7 kWh per sq. m per day (Ummadisingu and Soni, 2011). Solar photovoltaics power can effectively be harnessed providing huge scalability in India. Solar also provides the ability to generate power on a distributed basis and enables rapid capacity addition with short lead times. Off-grid decentralized and low- temperature applications will be advantageous from a rural electrification perspective and meet other energy needs for power and heating and cooling in both rural and urban areas. From an energy security perspective, solar is the most secure of all sources, since it is abundantly available. Theoretically, a small fraction of the total incident solar energy can meet the entire country's power requirements.

1.5 APPLICATIONS OF SOLAR ENERGY

Some of the applications of solar energy are:

1. Solar water heating
2. Solar heating of buildings
3. Solar electric power generation
4. Solar-pumping
5. Solar-distillation
6. Solar drying of agricultural and animal products
7. Solar furnaces
8. Solar cooking

1.6 SOLAR ENERGY COLLECTORS

The basic principle of solar thermal collection is that when solar radiation is incident on a surface (such as that of a black body), part of this radiation is absorbed, thus increasing the temperature of the surface. As the temperature of the body increases, the surface loses heat at an increasing rate to the surroundings. A Steady state is reached when the rate of solar heat gain is balanced by the rate of heat loss to the ambient surroundings. Solar concentrators increase the amount of incident energy on the absorber surface as compared to that on the concentrator aperture. The increase is

achieved using a reflecting surface or other means which concentrate the incident radiation onto a suitable absorber or receiver. Therefore, a solar concentrator generally consists mainly of (i) a focusing device, (ii) an absorber/receiver provided with or without a transparent cover, and (iii) a tracking device for continuously following the sun.

1.6.1 TYPES OF SOLAR THERMAL COLLECTORS

Based on the heating requirement and environmental conditions, three main variations of solar collector designs are:

1. Flat plate collector (FPC)
2. Evacuated tube collector (ETC) and
3. Concentrated solar power (CSP) collector

Based on their optical type, concentrating solar power technologies are classified as either point focus concentrators or linear focus concentrators.

Classification of linear focusing collectors are:

1. Parabolic trough collector (PTC)
2. Compound parabolic collectors (CPC) and
3. Linear Fresnel collector

Classification of point-focusing collectors are:

1. Parabolic dish reflector
2. Heliostat field collector

The solar flux concentration typically obtained is at the level of 30-100, 100-1000, and 1000-10000 for trough, tower, and dish systems; respectively.

1.6.1.1 Flat Plate Collectors

The flat-plate collectors (Fig. 1.4) are the most fundamental and the most studied technology in the field of solar thermal technology. The working of these collectors is very simple, it is just based on radiation passing through a transparent layer and then set on an absorber layer which absorbs the solar energy as heat energy. The absorbed heat is then transferred to a kind of medium fluid in the tubes to increase its temperature for direct thermal use.

As this collector is fixed in its position and doesn't track the sun it should be placed in such a way that the absorber layer faces south in the northern hemisphere and north in

the southern hemisphere. The optimum tilt angle of the collector is dependent on the attitude of the location with angle variations of 10-15 degrees. The efficiency of flat plate collectors is comparatively low, owing to their limited exit temperature range of about 50°C -140°C. This is because, as the collector temperature rises in the flat plate the heat loss to the surroundings will increase.

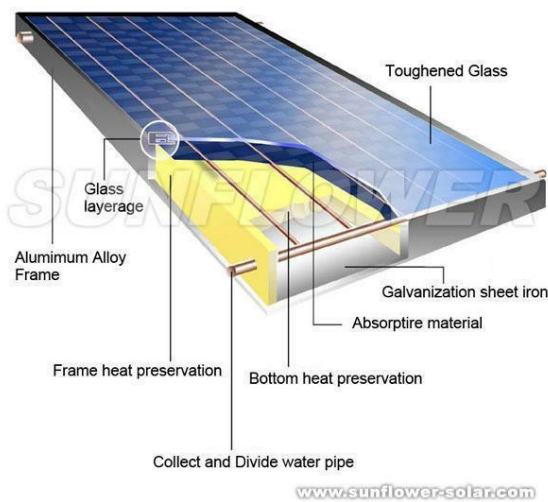


Fig. 1.4 Flat plate collector.

1.6.1.2 Compound Parabolic Collectors

These types of collectors (Fig. 1.5) can absorb virtually all the light emitted to their mouth of them. These can accept a large portion of diffuse radiations incident on their apertures and concentrate it at a point without tracking the sun. Like flat plate collectors, these collectors are fixed in a specific angle called acceptance angle. For stationary CPC collectors mounted in this mode, the minimum acceptance angle is equal to 47 degrees. This angle covers the declination of the sun from summer to winter solstices every year. This collector also has less efficiency as it cannot track the sun the entire day.

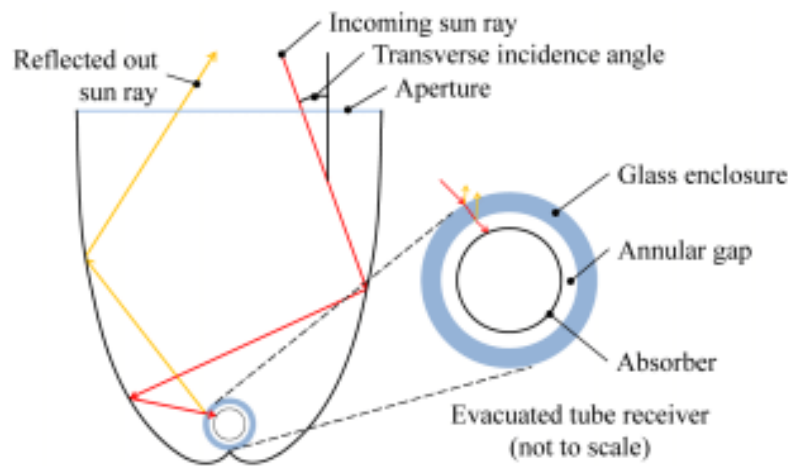


Fig. 1.5 Compound parabolic collectors

1.6.1.3 Evacuated Tube Collectors

Since the flat-plate collectors don't have a good optical performance in cloudy weather due to the condensation of moisture on the surface of the plate, evacuated heat pipe collectors were invented. As shown in Fig. 1.6, these collectors consist of a heat pipe inside a vacuum-sealed tube. The vacuum envelope reduces convection and conduction losses, so these collectors can operate to higher temperatures than flat-plate collectors. Evacuated tube solar collectors have significantly greater efficiencies due to the very high insulation power of the vacuum, where temperature can easily achieve up to 120°C to 180°C. Furthermore, the current advance of an evacuated flat plate can capture both beams and diffuse radiation and can achieve higher temperatures of around 200°C without concentration.

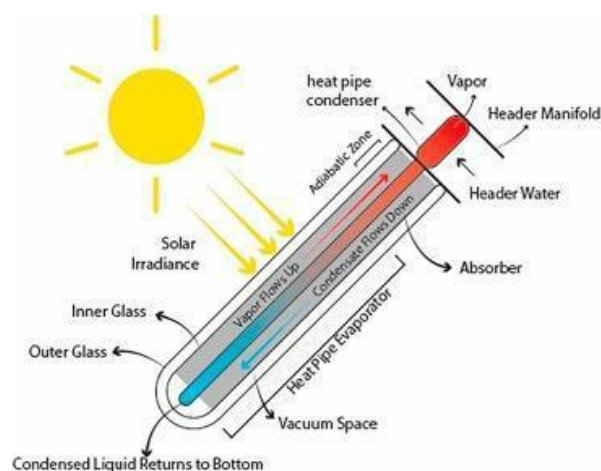


Fig. 1.6 Evacuated tube collector

1.6.1.4 Parabolic Trough Collectors

A parabolic trough collector (Fig. 1.7) can effectively produce heat up to temperatures of about 400°C (Desai and Bandyopadhyay, 2014). It consists of a parabolic mirror and a metal black tube covered with a glass tube to reduce heat losses. The tube is extended along the focal line of the mirror. When the parabola is pointed towards the sun, parallel rays incident on the reflector is reflected onto the receiver tube. It is sufficient to use a single-axis tracking of the sun to absorb the radiations. The collector can be oriented in an east-west direction, tracking the sun from north to south, or oriented in a north-south direction and tracking the sun from east to west.

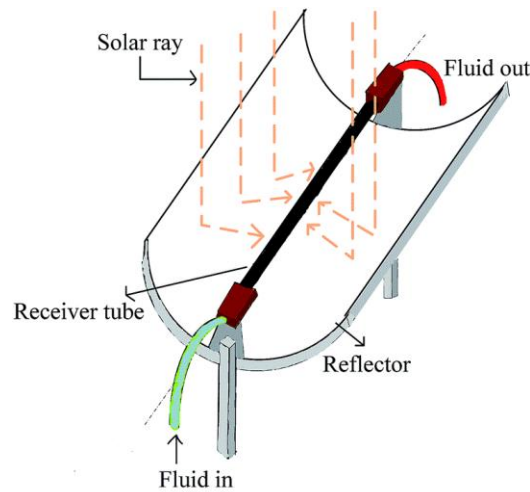


Fig. 1.7 Parabolic trough collector

1.6.1.5 Linear Fresnel Reflector

This type of collector (Fig. 1.8) utilizes flat or elastically curved reflectors which are cheaper compared to parabolic glass reflectors. Moreover, this type is mounted close to the ground, minimizing the structural requirements. Linear Fresnel collector is based on many mirrors focussing beams of light to the receiver tube which is surrounded by secondary reflectors. There are different designs for this kind of collector. In the normal design, there is just one absorber tube, but there are new designs that hold on to two or more tubes.

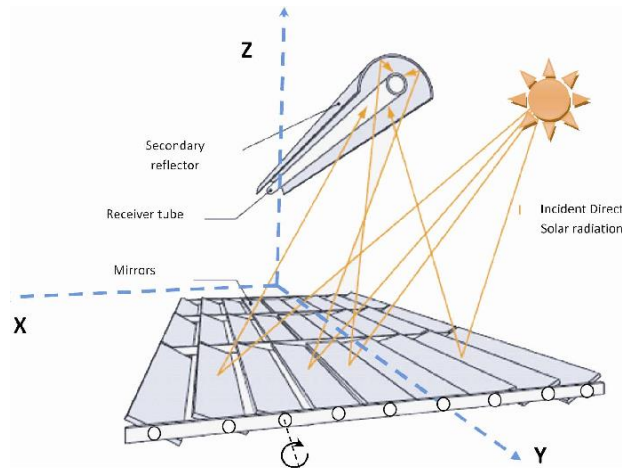


Fig. 1.8 Linear Fresnel reflector

1.6.1.6. Parabolic Dish Collector

The parabolic dish receiver (Fig. 1.9) tracks the sun in two axes during the day and focuses the sunlight irradiated to the dish surface to a focal point. The dish structure must follow the sun to reflect the beam to the thermal receiver. The receiver absorbs the radiant solar energy, converting it into thermal energy in a circulating fluid. These collectors can exceed the temperature of about 1500°C and they are more efficient than other non-tracking solar collectors (Kumar, Prakash and Dube, 2018).

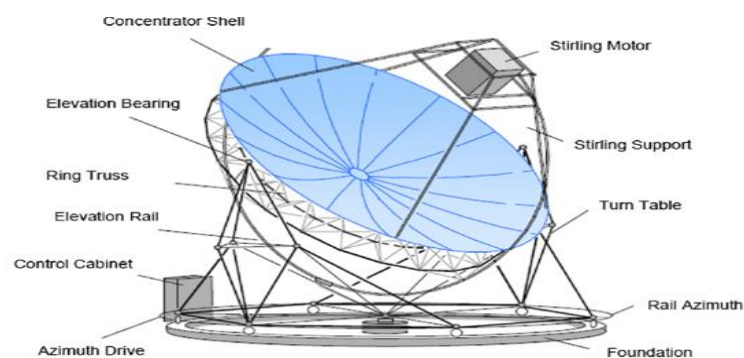


Fig. 1.9 Parabolic dish collector

1.6.1.7. Heliostat Field Collector

Heliostats are mainly concave tracking mirrors located around a tower in which there is a receiver on its head as shown in Fig. 1.10. In this, large amounts of solar energy would be concentrated in the cavity of the receiver, and this heat energy is transferred to

a circulating fluid that can be stored and later used to produce power. These collectors have a higher efficiency than non-tracking collectors.

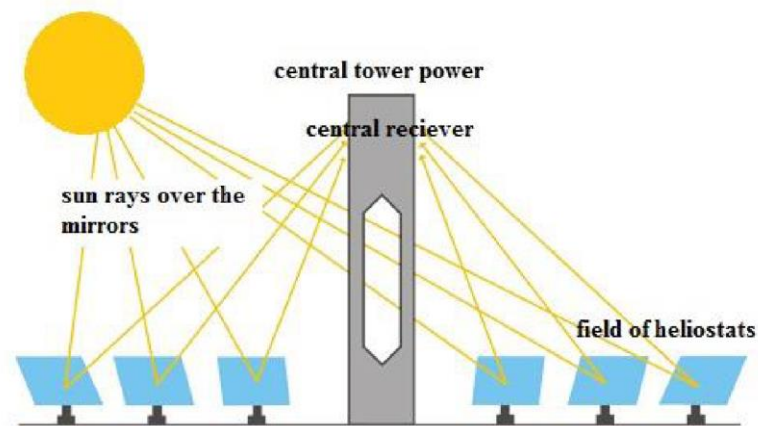


Fig. 1.10 Heliostat field collector

Among these solar power technological innovations, the parabolic trough collector is a hopeful area of research for high-efficiency solar power extraction and is widely used with different kinds of working fluids for medium/low-temperature solar energy utilization

1.7 SUN TRACKING SYSTEM

A Sun tracking system is the continuous positioning of the reflector towards the sun. As the sun's position changes hourly, the solar power device should be adjusted to produce the maximum output power. Regarding movement capability, two main types of sun trackers exist:

1. One axis tracker
2. Two axis tracker

The main difference among them is the ability to reduce the pointing error, increasing the daily irradiation incident to increase the energy output. The schematic block diagram of an active sun-tracking system is shown in Fig. 1.11.

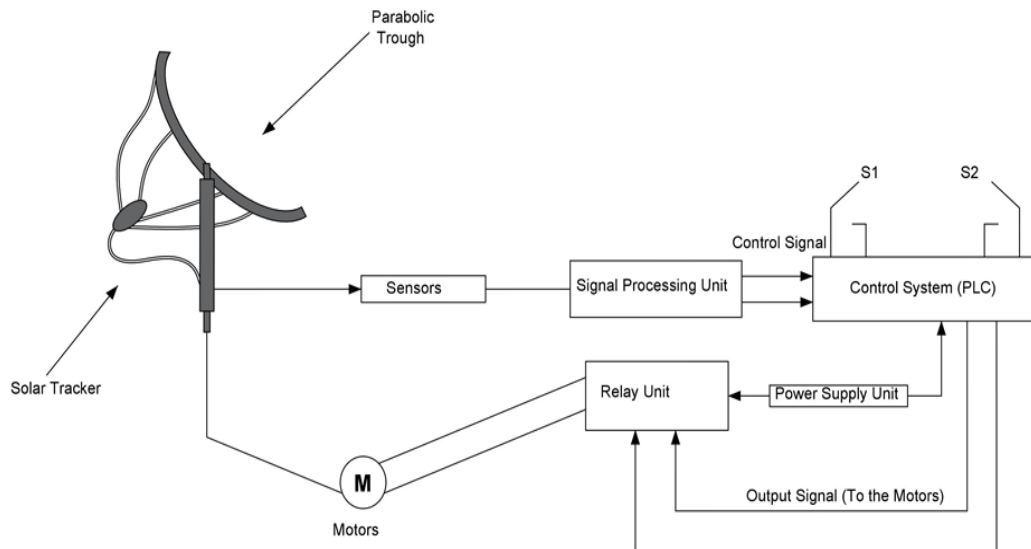


Fig. 1.11 Schematic diagram of an active sun-tracking system

Regarding control units, the main types of solar trackers are:

1. Passive
2. Microprocessors
3. Electro-optical controlled units

The first one doesn't use any electronic control devices or motors. The second type uses mathematical formula to predict the sun's movement while the third type uses information from some type of sensors to estimate the sun's position.

Regarding control strategy, the main types of solar trackers are:

1. Closed Loop
2. Open Loop
3. Hybrid System

The main difference among them is the time dependency and processing data flow. The first type is time-independent; it depends on the determination of the movement error aided by the solar sensor and feeds the sensor reading back to the actuators to correct the movement. The second type is time-dependent; it uses time and mathematical formulas to predict the sun's orientation. The hybrid system is the newest tracking system; it is a combination of the other two types, also the new algorithms i.e. Fuzzy logic, Neural Network, and neuro-fuzzy are listed under this category.

1.8 HEAT COLLECTION ELEMENT (HCE)

The heat collection element or parabolic trough solar collector receiver tube, as shown in Fig. 1.12, is located at the focus line of the parabolic reflector surface, with means of transferring the absorbed solar energy to a fluid. The HCE consists of the receiver tube that is surrounded by a glass envelope. The receiver is typically a stainless steel or copper tube coated with selective coating; this coating has high absorption of short-length solar radiation and low emissivity for the long-wave energy spectrum to reduce thermal radiation losses. The glass envelope is typically made from Pyrex (Eskin, 1999), which maintains good strength and transmittance under high temperatures. The outside glass envelope is transparent to solar radiation over the solar absorbing surface that reducing convection and radiation losses to the atmosphere; the annulus gap between receiver and glass envelope is under vacuum to reduce thermal losses.

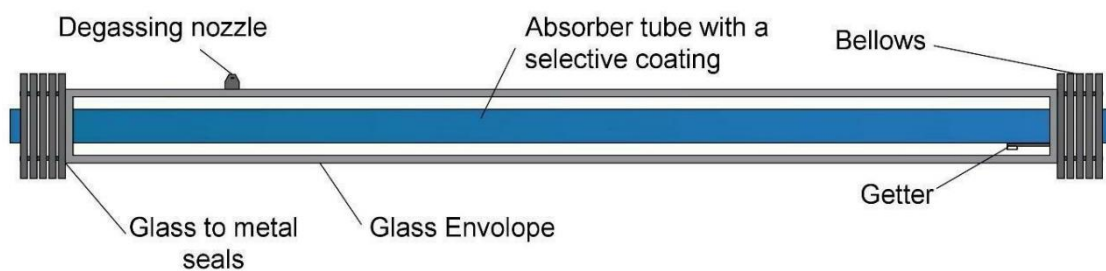


Fig. 1.12 Schematic of a receiver tube

1.9 HEAT TRANSFER FLUIDS OF PTSC

Heat transfer fluid carries heat from HCE to the heat storage tanks in solar heating (and cooling) systems. Typical working fluids include water, ethylene glycol, thermal oil, and air, while applications with molten salts (primarily nitrate salts) have been developed in the last years. When selecting a transfer fluid, the following criteria should be considered: the coefficient of expansion, viscosity, thermal capacity, freezing point, and boiling point. The following are some of the most commonly used heat transfer fluids and their properties.

1. Air: It will not freeze or boil and is non-corrosive. However, it has a very low heat capacity and tends to leak out of collectors, ducts, and dampers. The only

drawback of using air as a working fluid in the CSP system is the greater loss of pressure along with the collector fields, which leads to higher pump work.

2. Water: It is non-toxic and inexpensive. It has high specific heat and very low viscosity, making it easy to pump. Unfortunately, water has a relatively low boiling point and high freezing point. It can also be corrosive if the pH is not maintained at a neutral level.
3. Thermal oils: Oil-based thermal oils typically are categorized as either synthetic hydrocarbons, paraffinic hydrocarbons, or aromatic refined mineral oils. The major benefits promised by thermal oils include non-corrosiveness to most common metals, high thermal stability within the prescribed operating ranges, and, in most cases, little or no pressurization required even at high temperatures. These oils are a preferred choice for the temperature range between 200°C to 400°C, and the devices need not be pressurized to temperatures of up to 350°C.
4. Refrigerants/phase change fluids: These are commonly used as heat transfer fluids in refrigerators, air conditioners, and heat pumps. They generally have a low boiling point and a high heat capacity. Heat absorption occurs when the refrigerant boils (changes phase from liquid to gas) in the solar collector. Chlorofluorocarbon (CFC) refrigerants such as Freon, were the primary fluids used by refrigerators, air-conditioners, and heat pump manufacturers because they are non-flammable, low in toxicity, and stable, non-corrosive and do not freeze.
5. Molten salts: Molten salts are primarily ionic nitrate salts, used in high-temperature heat transfer applications because of their desirable properties at elevated temperatures. They remain in a liquid stage over a wide range of temperatures and are suitable as HTFs for high temperatures up to 550°C to 600°C (Bellos, Tzivanidis and Papadopoulos, 2018). High heat capacity, high thermal conductivity, high density, and small vapor pressure even at high temperatures are the unique characteristics of molten salts, which are essential features of a heat transfer/storage fluid.
6. Nanofluids: Nanofluids have recently found many important applications requiring quick and effective heat transfer. Moreover, in contrast to the conventional heat transfer fluids like water, hydrocarbon oils, refrigerants, molten salts, etc. nanofluids are not transparent to solar radiant energy; instead, they absorb and scatter significantly the solar irradiance passing through them. Several

investigators have reported substantial improvements in the thermal gain rate with the use of several nanofluids in solar applications compared with traditional fluids (S. Shajan and V. Baiju, 2020). Nanofluids have suspended nanoparticles suspended in them and these particles enhance the thermal conductivity which results in an improvement in the efficiency of the heat transfer system than the conventional fluids.

1.10 STORAGE OF THERMAL ENERGY

Thermal energy may be stored in the forms of 1) sensible heat, 2) latent heat. These methods differ in the amount of heat that can be stored per unit weight or volume of storage media and operating temperatures.

1. **Sensible Heat Storage:** In sensible heat storage, the thermal energy is stored by changing the temperature of the storage medium. The amount of energy required to raise the temperature of one unit of a material one degree is its specific heat, however, the amount of heat stored depends on the heat capacity of the media being used, the temperature change, and the amount of storage media. Water and pebbles are the most common materials used for low-temperature energy storage and hydrocarbon oil has been proposed for high temperature.
2. **Latent Heat Storage:** In latent heat storage, thermal energy is stored using a reversible change of state (phase change) in the media. Solid-liquid transformations are most commonly utilized. Liquid-gas and solid-gas phase changes involve the most energy of the possible latent heat storage methods. A common media used for latent heat storage in solar thermal systems is molten salt.

1.11 OBJECTIVES

The broad objective of the project is:

- To design a PTC with a secondary reflector for uniform heat flux distribution and desired output power

The specific objectives of the project are:

1. To analyze the optical performance and power output of the PTC by numerical solution.
2. To evaluate the influence of a secondary reflector for the PTC to homogenize the solar heat flux on the receiver.
3. To select and fabricate the desirable secondary reflector with the PTC.
4. To conduct an experimental study to evaluate the performance of the modified PTC.
5. Performance prediction modelling using Machine learning algorithms

1.12 SCOPE OF THE STUDY

Solar power is an unlimited source of energy which is an alternative to fossil fuels as it is a non-polluting, abundant, reliable, and renewable source of energy. Hence the utilization of this energy is important. This can be used to generate electricity and meet other demands of heating, cooling, soil disinfection, and domestic hot water requirements (Gulzar, Qayoum and Gupta, 2019). The solar thermal collector is a technology to harness this energy. The main drawbacks of the parabolic solar collectors are the temperature limitations of the heat transfer fluids, low thermal energy conversion efficiency, and structural failure due to uneven heat flux distribution.

The project aims to develop a PTC which is more efficient than the conventional one. And, to design a parabolic trough for solar collector with a secondary reflector of optimum shape. The modified PTC is highly efficient, has uniform stress distribution, low tube wear - due to the non-uniformity of the temperature distribution over the circumferential angle of the tube and has maximum power output.

1.13 REPORT OUTLINE

Chapter 1 provides a general introduction. Chapter 2 deals with the background information on the approaches and studies related to the development and characterization of the secondary reflector and also on the design and analysis of parabolic troughs in the solar collector. Chapter 3 discusses the methodology for the experiment. Chapter 4 provides details on the component of the collector and also provides information on how the collector is designed. Chapter 5 deals with the optical simulation of the newly designed parabolic trough, and numerical analysis. Chapter 6 deals with the design optimization and experimentation of secondary reflector. Chapter 7 deals with experimentation and chapter 8 deals with machine learning of the PTC with selected shape of secondary reflector. Finally, Chapter 9 documents the conclusion obtained from the experiment.

CHAPTER 2: LITERATURE REVIEW

The purpose of this chapter is to provide a literature survey of past research efforts such as journals or articles related to the design of the composite parabolic trough collector and its efficiencies

1. Brooks M J et al. (2005) fabricated a parabolic trough collector with water as a working fluid. Both an evacuated glass shielded receiver and unshielded receiver were tested with water as the, with which peak thermal efficiencies of 53.8% and 55.2% were obtained respectively. The acceptance angles for the unshielded and glass-shielded receivers were similar.
2. F.J. Cabrera et al. (2013) developed a modular, small, lightweight, and low-cost parabolic collector, compatible for installation on the roofs of the buildings aiming to overcome some of the current technology drawbacks such as costs and modularity. Applications of PTC in solar cooling systems with potential application in feeding double effect absorption chillers.
3. Himangshu Bhowmik & Ruhul Amin developed a flat plate absorber tube of iron and studied the efficiency of the system with and without a collector. The results showed that the efficiency of the system increases by 10% when using a reflector material.
4. Soteris A. Kalogirou. et al. (1994) describe a low-cost method for mass-production of parabolic surfaces with fiberglass. The cavities produced with plastic conduits, covered with fiberglass at the back of the collector surface, reinforce the longitudinal and transverse directions, to increase rigidity.
5. Soteris A. Kalogirou (2012) conducted the thermal analysis of the collector receiver taking into consideration all modes of heat transfer.

6. Thomas (1994) describes a simple welded frame structure for parabolic trough concentrators suitable for developing countries. He conducted static load tests on this structure to study its deflection characteristics under various load conditions. The results show that the slope error of the reflector support corresponding to normal wind loads was found to be within the specified limits, and the structure withstood the load corresponding to extreme wind load conditions.
7. Thomas et al. (2009) reviewed the design concepts of structural, optical, and thermal subsystems of parabolic trough concentrators. Existing methods of performance evaluation and techniques to improve their performance were also studied.
8. V. Pranesh. et al. (2019) conducted a comprehensive review of the CPC solar collector was carried out for the past 50 years in terms of types, historical growth in milestones, concept, design strategies, heat transfer fluids, experimental studies, theoretical studies, applications, standards, certifications, market players and recent developments.
9. Prakash et. al. (2017) tested two different reflectors for performance analysis of PTC. Stainless steel sheets and acrylic mirror sheets are used as reflectors. S. S. reflector reported higher efficiency than an acrylic reflector.
10. Folaranmi J. et al. (2009) developed the design, construction, and testing of a Parabolic Solar Steam Generator collector with a solar tracking system unit. The system was evaluated only based on manual tracking and the test results gave high temperatures above 200°C on average sunny and cloud-free days.
11. Ahmet Caglar et al. (2016) developed the design of a Parabolic Trough Solar Collector using a Concentrator with high reflectivity by comparing fluid temperatures while using two different reflective surfaces, a chrome plate, and a composite aluminum panel. The results show that composite aluminum panel reflector has a better performance than chrome plate reflector due to high reflectivity.

12. Alka Bharti et al. (2019) conducted a thermal performance analysis of a small-sized solar parabolic trough collector with and without secondary reflectors. The circumferential temperature distribution on the outer surface of the receiver tube is obtained more uniform in the case of PTC with secondary reflectors than in the case without secondary reflectors.
13. Huang et al. (2017) investigated different absorber tube modifications using dimples, protrusions, and helical fins and proved that dimpled tubes showed much better results for improving the thermal efficiency of parabolic trough absorbers.
14. Ziwei Zhao et al. (2020) compared a smooth receiver tube and pin finned receiver tubes for a parabolic trough solar air collector. Energy efficiency and energy efficiency in the pin finned tubes was found to be 10.4–14.5% and 2.55–4.29% higher than those of the smooth tube, respectively.
15. Ghasemi and Ranjbar (2017) found that the inclusion of porous rings within the receiver tubes impressively improves the heat transfer coefficient, while Bellows and Tzivanidis(2018) used geometry modification inserts with star shape that revealed an increase in the thermal efficiency.
16. Nain et al. (2021) investigated the performance of different U-shaped heat exchangers of copper and aluminum with and without fins alternately in parabolic trough solar collectors for air heating applications. They observed that the system performance with the copper heat exchanger is better than the aluminum heat exchanger. At elevated temperatures, the percentage increase in solar air heater performance for the aluminum heat exchanger is higher than for the copper heat exchanger.
17. Seyed Ebrahim Ghasemiet al. (2017) studied the effect of heat transfer nanofluid flow inside the receiver tube of a parabolic trough collector by numerical simulation. The results show that heat transfer and thermal performance of parabolic solar collectors using nanofluid were improved as compared to the base fluid.

18. Norouzi et al. (2020) conducted the efficiency enhancement of parabolic trough solar collector using the rotating absorber tube and nanoparticles. The results show that aluminum is the best material for absorber tubes compared to other materials and also a more uniform surface temperature distribution and a higher collector thermal efficiency could be met.
19. Haddouche Mohammed Reda et al. (2019) moved the absorber tube downward away from the focal line of the parabola and a secondary reflector is added overhead the tube to reduce the heat flux gradient and homogenize it
20. Evangelos Bellos et al. (2018) proved from the thermal analysis that the thermal enhancement is greater than the optical and it is higher when the system operates with great fluid temperature levels. the use of the extra booster leads always to optical and thermal enhancement, especially in greater incident angles. The yearly optical enhancement, for a collector with a focal distance to length ratio of 0.236, is calculated at 21.7% and it is more intense in the months with greater incident angles.
21. Alka Bharti et al. (2019) represented the theoretical analysis of receiver tube misalignment, the design of secondary reflectors, and experimental analysis of a small-sized solar parabolic trough collector (PTC) with and without secondary reflectors. Experimental analysis of PTC has been done using a parabolic secondary reflector (PSR) and triangular secondary reflector (TSR) and compared with PTC without a secondary reflector (WSR).
22. WANG Kun et al. (2014) analyzed the solar flux distribution on the outer surface of the absorber tube, the temperature field of the absorber tube wall, and the collector efficiency. It was revealed that the absorber tube could almost be heated uniformly in the PTSC with an HR. As a result, the circumferential temperature difference and the maximum temperature could be reduced significantly, while the efficiency tended to decrease slightly due to the inevitably increased optical loss.

23. R. Abbas et al. (2015) obtained the optical annual energy and energy efficiencies for PTCs and LFCs, with a multitude of secondary reflector receivers, where the concentration at each tube is used to weigh the exergy efficiency. It results that the maximum efficiency is obtained by PTCs, while the lowest one corresponds to LFCs with a multitube receiver. Also, it is concluded that, while for PTCs NS orientation leads to clearly higher efficiencies, this is not the case for LFCs, where both orientations achieve similar efficiencies even when the solar field has been designed for NS orientation.

CHAPTER 3: METHODOLOGY

First, a thorough literature survey has been carried out, which included looking at research articles from respected international journals and publications. Based on the literature review the work is conducted in a flow as represented in the flow chart as shown in Fig.3.

Optical simulation of the PTC is done using numerical methods through a comprehensive literature review to analyze different types of secondary reflectors and to evaluate heat flux distribution over the receiver tube. Power output has been determined from the solar collector. Tonatiuh an open-source, freeware, Monte Carlo ray tracing software is used for the simulation where we can visualize the radiative flux distribution of the modelled surface. The geometries of different secondary reflector designed in Solidworks has been imported to Tonatiuh. Power output of the solar collector and heat flux distribution over the circumference of the receiver tube has been determined for the different secondary reflector considered. Fabrication of the optimum secondary reflector and performance of the solar collector, with and without a secondary reflector has been evaluated for actual weather conditions through experimentation. Finally, performance prediction modelling using machine learning techniques.

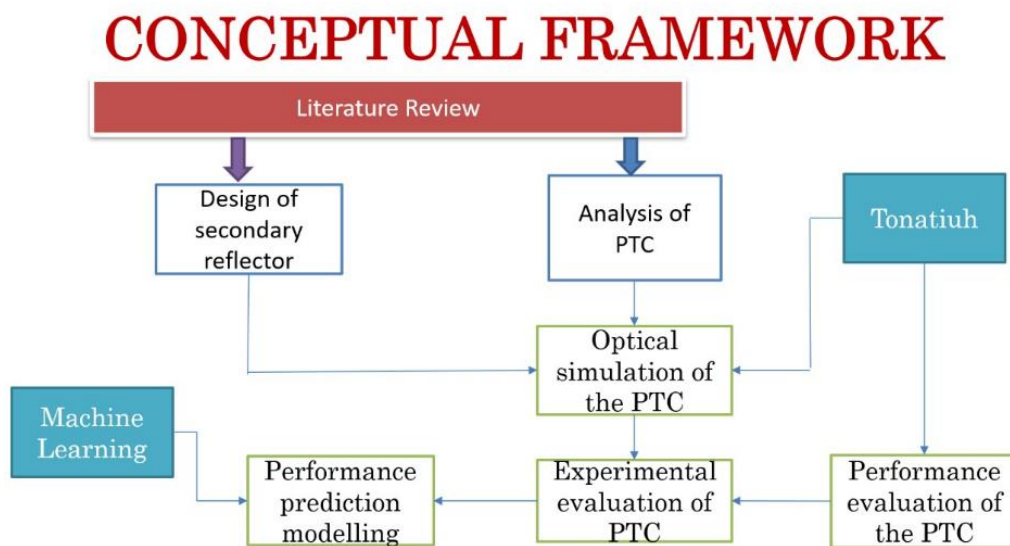


Fig 3. Conceptual framework of the study

CHAPTER 4: DESIGN OF A PARABOLIC TROUGH COLLECTOR

In this chapter Parabolic Trough Solar Concentrator is designed and all its dimensions are determined.

4.1 BASIC COMPONENTS OF A PARABOLIC TROUGH COLLECTOR

A solar collector is a special kind of heat exchanger that converts solar radiation energy into heat. In solar collectors, energy transfer is from a distant source of radiant energy to a fluid. The basic components of a solar parabolic trough collector system consist of the following parts:

4.1.1 Collector

The collector is one of the vital parts of the parabolic trough solar collector system, as it decides the fraction of solar insolation to be collected by the absorber tube. A parabolic collector reflects and concentrates all the sun rays onto the absorber tube. The collector is parabolic shaped, normally made of galvanized aluminium or stainless-steel sheet.

4.1.2 Absorber Tube

The absorber tube is placed along the focal length of the parabolic trough collector. The solar radiations reflected by the parabolic trough collector are collected by the absorber tube. Normally used absorber tubes are mild steel, aluminium, and copper. The working fluid in the absorber tube may be air, water, or any heat transfer oil.

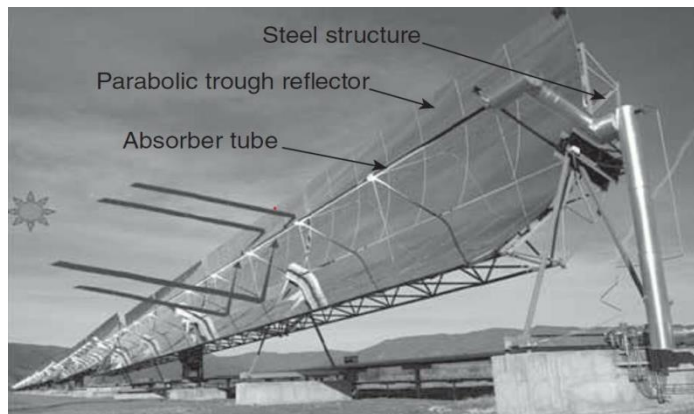


Fig. 4.1 A typical parabolic trough collector (Moya et al., 2012)

4.2 DESIGN OF PARABOLIC TROUGH COLLECTORS

The primary objective in this collector's design and construction was to demonstrate that a stable, effective parabolic profile was ensured on the reflective surface. This should be achieved if the deflection of the trough under severe wind load is negligible. The geometrical parameter of a parabolic trough solar collector is shown in the Fig.4.2.

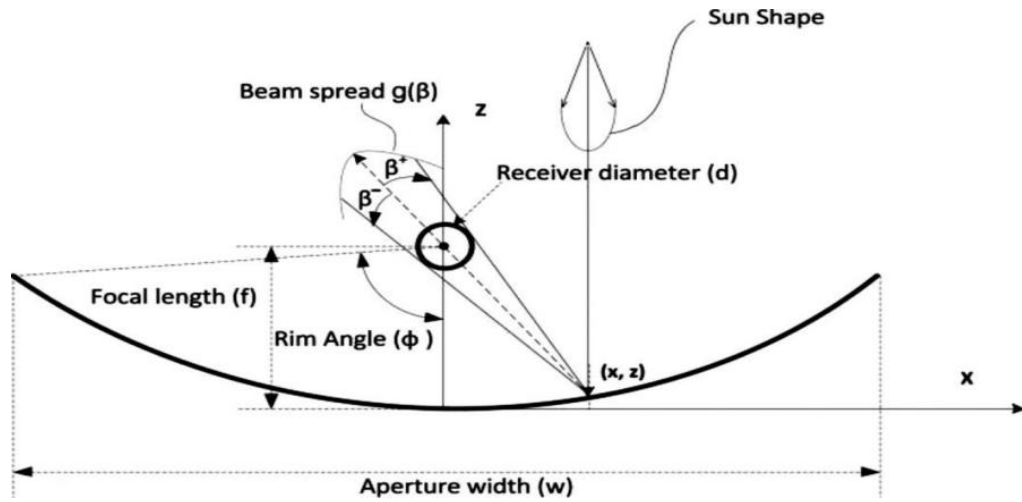


Fig. 4.2 Nomenclature of parabolic trough collector

The purpose of this study is to design a PTC, which is intended for the driving source of the refrigeration system, which has a performance coefficient of 4.5 (COP) and a power of 5500 W (refrigeration effect). The geometric parameters of the PTC are obtained by the basic design procedure as given below.

For the refrigeration system,

$$COP = \frac{\text{Refrigeration effect}}{\text{Heat input}} = \frac{RE}{HI_R} \quad 4.1$$

On solving equation 4.1 we get the required heat input (HI_R) to the cooling system as 1222W. The energy required by the cooling system (HI_R) is supplied from the output power of the PTC system (OP_{PTC}).

$$HI_R = OP_{PTC} = 1222 \text{ W} \quad 4.2$$

By assuming an efficiency of PTC as 30% ($\eta = 0.3$) by considering all errors and losses.

$$\eta_{PTC} = \frac{OP_{PTC}}{IP_{PTC}} \quad 4.3$$

$$IP_{PTC} = \frac{1222}{0.3} = 4073 \text{ W} \quad 4.4$$

$$IP_{ptc} = I * A \quad 4.5$$

Assuming an average solar intensity (I) as $600 \frac{W}{m^2}$, On solving equation 4.4 we obtain aperture area (A) of the PTC.

$$A = \frac{4073}{600} = 6.78 \text{ m}^2 \approx 7 \text{ m}^2$$

The Concentration ratio of the PTC is given by

$$C_c = \frac{W_a}{\pi D_o} \quad 4.7$$

Assuming Concentration ratio $C_c = 33$ and diameter of absorber tube (D_o) as 0.02665 m, on solving equation 4.5 we obtain the aperture width (W_a) as 2.76 m $\approx$ 2.8 m.

The area of the PTC is given by

$$A = W_a \cdot L \quad 4.8$$

Focal point (f) is the distance between the vertexes of a parabola, which defines the parabola completely

From Cartesian coordinates

$$x^2 = 4fy \quad 4.9$$

Aperture width

$$W_a = 4 \times f \times \tan \tan \left(\frac{\varphi}{2} \right) \quad 4.10$$

We take rim angle $\varphi = 70^\circ$ in order to minimise the maximum distance between the parabolic collector and the focus.

We have $\varphi = 70^\circ$ and $W_a = 2.8m$, on solving equation 4.7 we get the focal length of the collector as 0.425m.

$$f = 1.0 \text{ m}$$

The dimensions of the parabolic trough collector are stated in Table 4.1.

Table 4.1: Dimensions of PTC

Parameter	Value
Parabolic Length	2.5m
Parabolic Aperture	2.8m
Focal Length	1 m
Aperture area	7 m^2
Rim Angle	70°
Diameter of the receiver (D)	0.02665m

CHAPTER 5: OPTICAL SIMULATION

5.1 INTRODUCTION

In conventional parabolic trough solar thermal collectors, the concentrated solar radiation distribution on the circumference of the receiver tube is highly non-uniform, which is one of the primary causes of receiver failures. A secondary reflector placed somewhere near the receiver tube can be used to homogenize the solar heat flux distribution over the circumference of the receiver tube (Bharti, Mishra, and Paul, 2019). However, the location of the receiver tube and the secondary reflector configuration may significantly influence the performance of the system. Therefore, this study examines the influence of the position of the receiver tube and the geometry of the secondary reflector on application of a secondary parabolic reflector for homogenizing the concentrated solar flux distribution on the receiver tube to achieve a uniform heat flux distribution over the receiver tube surface.

5.2 SECONDARY REFLECTOR

The conventional parabolic trough collector is designed to capture the direct solar radiation over a large parabolic-shaped surface and concentrate it onto its focal line. The concentrator is a sheet metal bent to a parabolic shape and painted with a reflective coating to reflect the solar irradiation on its focal line. The absorber tube is the major component of PTC, in which solar radiation is focused and converted to thermal energy by an intermediate of a heat transfer fluid (HTF). From Fig. 5.1a, the absorber tube is subjected to a non-uniform heat flux while the bottom periphery is subjected to concentrated solar radiation and the top one is subjected to non-concentrated solar radiation. The non-uniformity of the solar flux distribution over the outer surface of the absorber tube leads to a large difference in the temperature distribution which can cause damage to the structure of the absorber tube (Wang *et al.*, 2015). The induced stresses due to this uneven heat flux make the receiver tube deviate from the focal axis of the solar concentrator, thus rising optical losses.

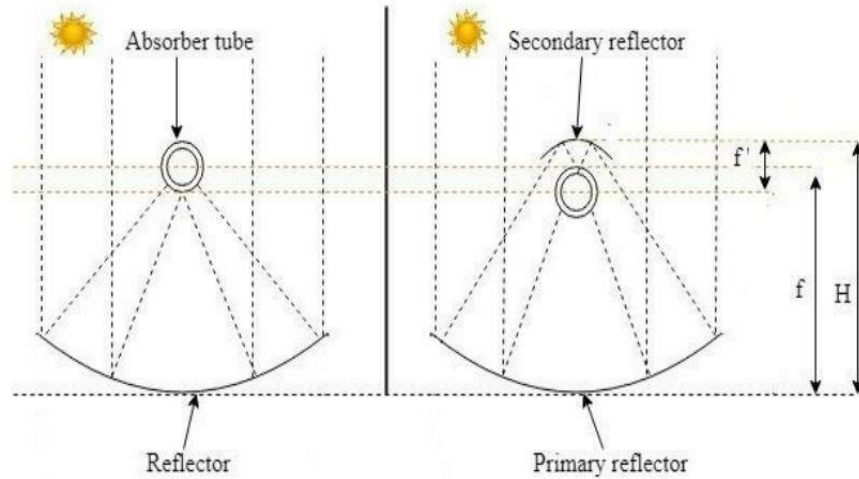


Fig. 5.1 The schematic diagram of the parabolic trough collector

One of the effective ways to homogenize the heat flux and reduce the circumferential temperature gradient is to move the absorber tube away from the focal line of the parabolic trough concentrator toward the concentrator and a secondary reflector is added overhead of the absorber tube. The two parabolas are arranged oppositely for uniform heat flux distribution on the tube. The sun rays reflected by the primary concentrator hit the bottom part of the absorber tube and a portion of these rays are reflected again on the upper part of the absorber tube using a secondary reflector as shown in Fig. 5.1b. Therefore, this study investigates the application of a novel secondary parabolic reflector for homogenizing the concentrated solar flux distribution on the receiver tube to achieve a uniform flux distribution over the receiver tube surface. The influence of receiver tube and SR locations on heat flux distributions and output power of the system was examined using Monte Carlo ray tracing (MCRT). Tonatiuh software has been used to simulate and estimate the distribution of solar flux across the surface of the absorber tube and the output power of the PTC in various configurations.

5.3 MONTE CARLO RAY TRACING (MCRT) TECHNIQUE

Monte Carlo Ray Tracing (MCRT) method is a powerful tool for simulating the optical performance of the parabolic trough collector (Wang *et al.*, 2015). This method

combines the Monte Carlo Method (MCM), based on statistics and randomized trials, with the Ray Tracing Method (RTM), based on geometrical optics. The basic idea of MCRT is as follows: -

At first, the position and direction of the incident rays coming from the sun are initialized according to the original sun shape model. And then, the optical behaviours, such as reflection, transmission, and absorption, on each interface are determined by a series of uniformly generated random numbers, and the propagating path of each ray is traced accordingly. Finally, the landing positions of the rays on the outer surface of the absorber tube are recorded. From the recorded data, the energy flux density on the absorber surface is determined, and some other optical performance can also be further obtained.

5.4 RAY TRACING TOOLS

The most used ray-tracing tools are SolTRACE and Tonatiuh. SolTRACE is a new optical modelling software to model solar power optical systems and analyze their performance (Wendelin, Dobos, and Lewandowski, 2013). It is used to model and predict the performance of new and complex solar optical designs that were not able to be achieved earlier. It is used specifically for the Windows 2000 operating environment. The ray-tracing methods associated with the speed and memory capabilities of the Windows 2000 operating environment provide more accurate and rapid results. The software uses a ray-tracing methodology and each ray selected by the user is traced through the system. The main advantage of this tool over the earlier ones is the high accuracy for complex systems.

Tonatiuh is an accurate and easy-to-use Monte Carlo ray tracer for the optical simulation of solar concentrating systems (Wang *et al.*, 2018). It advances the simulation tools available for the design and analysis of solar concentrating systems. The main aim of the software is to develop a strong theoretical base that facilitates the optical simulation of almost any type of solar concentrating system. It provides a clean and flexible software architecture that allows the user to enhance and modify its

functionalities with ease. For performing the optical simulation of the parabolic trough solar thermal collector, Tonatiuh software is being used.

5.5 PARAMETERS OF PTC

Figure 5.2 shows a schematic representation of the PTC with a parabolic SR. Solar radiation reflected by the primary reflector are focused on the bottom periphery of the absorber while the rays that deviate from the tube are re-reflected by the secondary reflector on the top surface of the tube. The geometric parameters of the main collector and receiver tube are presented in the Table. 5.1.

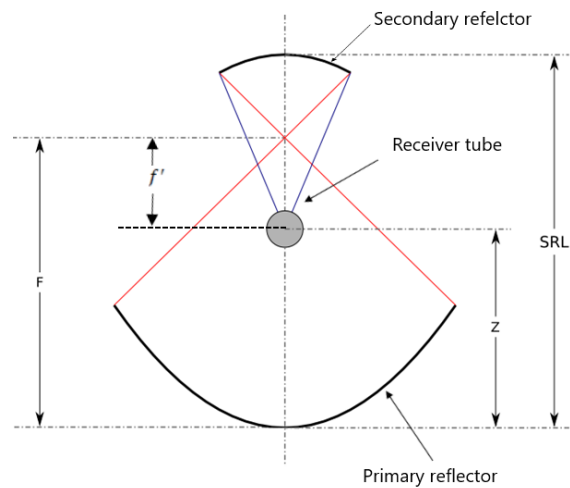


Fig. 5.2 Schematic diagram of PTC with secondary reflector

To study the downward misalignment (z) of the receiver tube position on the heat flux distribution the tube is placed 0.95m from the primary collector. Similarly, 1.05m is considered for the upward misalignment (z) of the receiver. For the above examples, the secondary reflector positions (SRL) are adjusted at 1.05m and 1.15m, respectively, such that the secondary reflector's focal axis is aligned with the tube's axis.

Table 5.1 Parameters of PTC

Item	Value
Aperture area of the collector	7 (m ²)
Aperture width (Wa)	2.8 (m)
Collector length (L)	2.5 (m)
Focal distance (f)	1 (m)
Rim angle	70°
Diameter of the receiver (D)	0.02665 (m)
Geometric concentration ratio	33
Width of secondary reflector	0.06 (m)

5.6 OPTICAL MODEL OF PTC

The optical models of PTC with different absorber tube positions have been carried out using MCRT simulations. The optical simulation is performed with Tonatiuh, a ray-tracing software. The Blue sun shape model which is considered the most accurate is

used for defining the solar irradiance. According to this model, the distribution of terrestrial solar flux is given, as shown in Eq. (4.1) (Chen, Li, and Zhang, 2015).

$$L^{\wedge}_{Biue}(\theta) = \begin{cases} \frac{\cos 326(\theta)}{\cos 308(\theta)}, & 0 \leq \theta \leq \theta_{disk} e^{\kappa(10^3\theta)^{\gamma}}, \\ & \theta_{disk} < \theta \leq \theta_{aureole} \end{cases} \quad (4.1)$$

where ϕ is the radial angular displacement, γ and χ are power functions of the circumsolar ratio.

The annual averaged angular width of the solar disk, $\phi_{disc} \approx 4.65 \times 10^{-3}$ radians, and the angular extent of the aureole, $\phi_{aureole} \approx 43.6 \times 10^{-3}$ radians (Wang *et al.*, 2020). In this analysis, the reflectance of the reflectors was taken as 0.98 and the absorptance of the absorber as 0.98. Beam irradiation of 1000 W/m² and a circumsolar ratio of 0.02 was considered for the simulation. With this model, the effect of geometric variations, such as the receiver tube position and secondary reflector geometry on the heat flux distribution and power output can be studied.

5.7 RESULTS AND DISCUSSION

Using Tonatiuh software, the uniformity of heat flux distribution on the circumference of the receiver tube and the power output of the PTC with an SR was numerically studied. The observations are shown for two options: (i) PTC with the receiver tube below the focal axis, and (ii) PTC with the receiver tube above the focal axis.

5.7.1 Ray Tracing simulation

A simulation with Tonatiuh has been conducted to calculate the distribution of solar heat flux around the absorber tube. The sun shape is set as a Buie model with two million numbers of rays considered in optical modelling, however, 1000 rays are selected to display a clear representation of the ray path, as illustrated in Figs. 5.3a and 5.3b. When the receiver tube is situated on the proper focal axis, as illustrated in Fig. 5.3a, the whole reflected radiation is reflected into the tube's bottom periphery, and the receiver is subjected to uneven distribution of solar radiation flux. When the receiver is

moved away from the focal axis and a secondary reflector is placed over the tube, the whole periphery is exposed to focused radiations, as shown in Fig. 5.3b, resulting in a more uniform distribution of solar radiation flux.

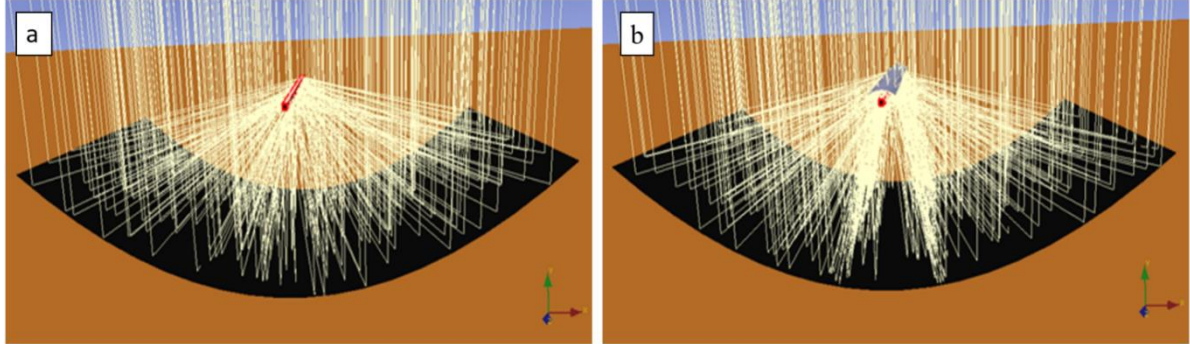


Fig. 5.3 Traced paths of rays (a) without reflector, (b) with reflector

5.7.2 Heat flux

Heat flux distribution on the surface of the absorber tube, when the tube is placed at different locations, such as at the focal axis, above the focal axis, and below the focal axis, is shown in Fig. 5.4. Here a parabolic geometry is used as the secondary reflector. For the three evaluated configurations, the maximum local heat flux observed are 70000 W/m², 60000 W/m², and 32000 W/m², respectively, for the PTC with the secondary reflector at the focal axis, above the focal axis, and below the focal axis, respectively. The local hot pocket for the PTC with the secondary reflector placed below the focal axis is significantly smaller than the other two configurations. This smaller local heat flux value is due to the improved solar flux distribution with the help of the secondary reflector. Hence, it can be concluded that applying the secondary reflector when the receiver is mounted below the focal axis of the primary collector is beneficial for the uniform heat flux distribution. However, the position of the receiver tube, geometry and position of the secondary reflector are to be determined for a targeted output power with a better uniform heat flux distribution.

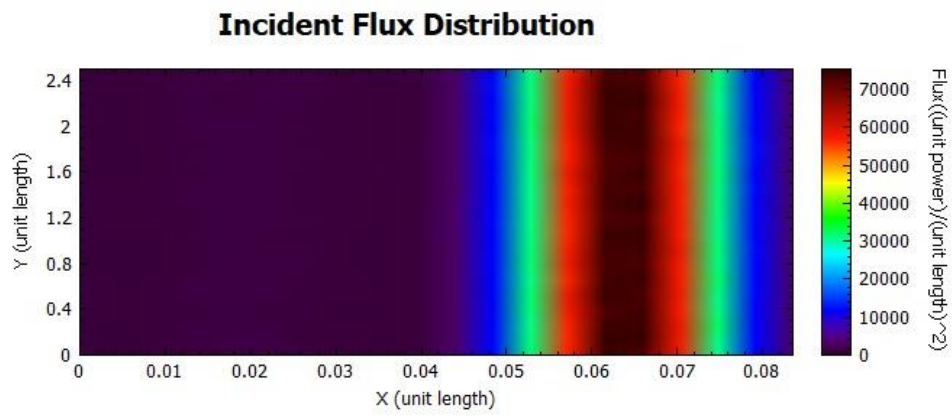
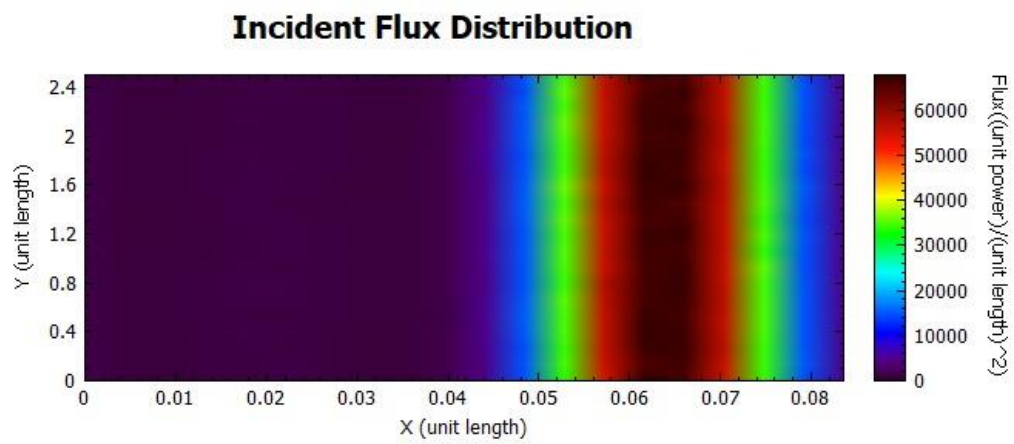


Fig. 5.4a Flux distribution on absorber surface at the focal axis



5.4b Flux distribution on absorber surface above the focal axis

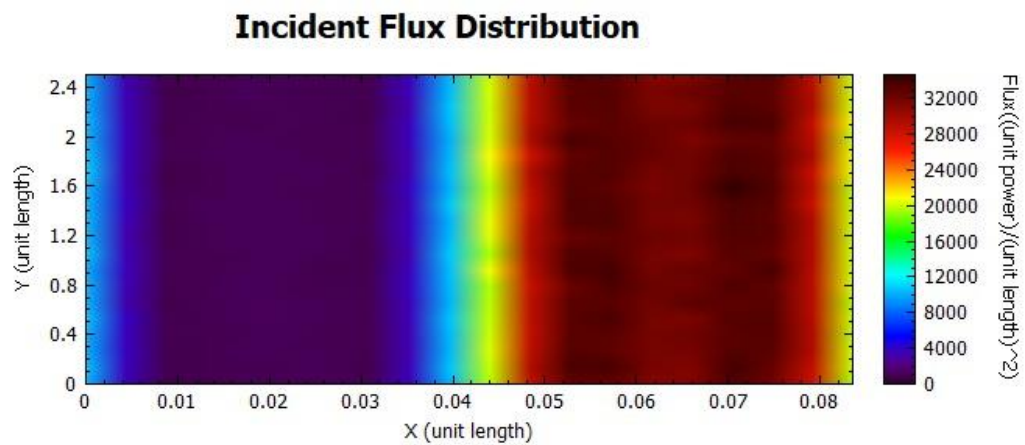


Fig. 5.4 Flux distribution on absorber surface (a) at the focal axis, (b) above the focal axis, (c) below the focal axis

The result shows that when the receiver tube is at the actual focal axis, the heat flux distribution is highly non-uniform, as shown in Fig. 5.4b and 5.6. The simulation result indicates that the heat flux distribution is highly non-uniform when the receiver tube is at the real focal axis, as shown in Fig. 5a and similar results can be seen in the other investigations related to PTC receivers (Bharti 2019, Mishra and Paul, 2019). A very large intensity of heat can be obtained, but it is mainly concentrated only in a small portion of the receiver tube. This results in a non-uniformity across the surface and further leads to severe tube distortion. This ultimately leads to a decrease in the optical efficiency of the system. The peak heat flux, in this case, is on the order of 67500 W/m^2 , and the flux gradient between the bottom and upper areas of the receiver surface is significant, as shown in Fig 5.4b. Fig 5.4a and Fig 5.5 shows the heat flux distribution and the graphical representation, respectively, when the receiver tube is placed above the focal axis. The majority of the reflected radiations are focused on a small region of the tube's bottom surface. This results in a highly uneven heat flux distribution with a severe peak heat flux of the order of 73600 W/m^2 , further leading to extremely severe distortions across the tube (Fig.5.4a).

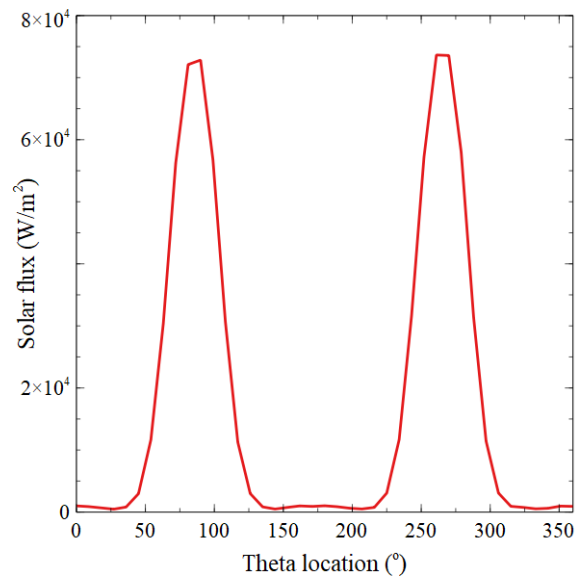


Fig. 5.5 Heat flux distribution for receiver above the focal axis

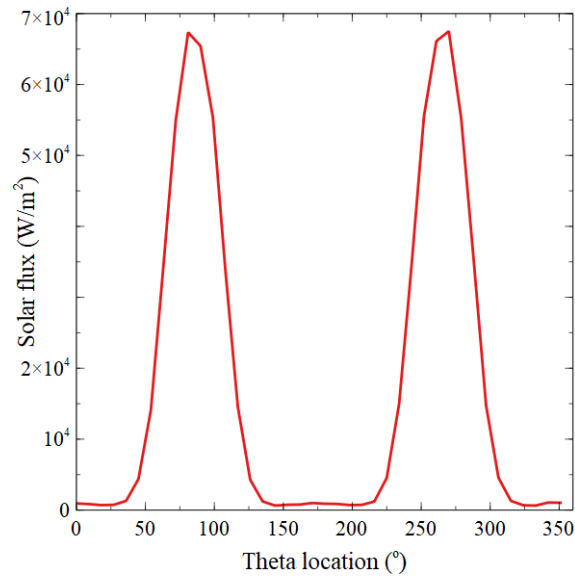


Fig. 5.6 Heat flux distribution for a receiver at the focal axis

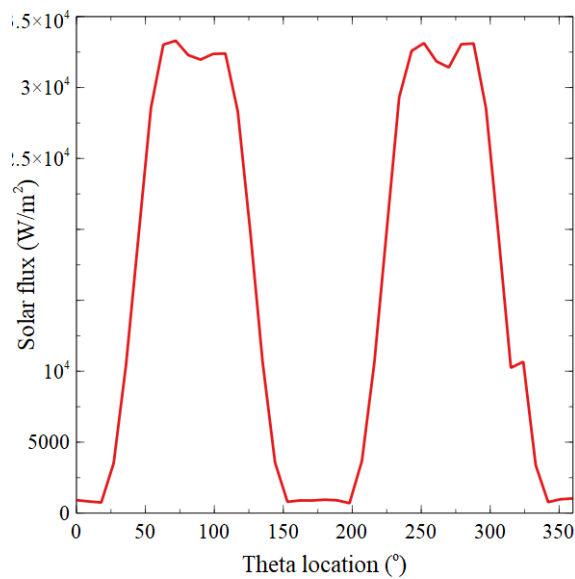


Fig. 5.7 Heat flux distribution for the receiver below the focal axis

Fig 5.4c and 5.7 shows the heat flux distribution and the graphical representation, respectively, when the receiver tube is placed below the focal axis. The part of the radiation that could not intercept the receiver tube can be re-reflected on the top circumference of the tube by installing a secondary reflector. As a result, the heat flux distribution throughout the receiver becomes more homogenous with a reduced flux gradient (peak heat flux 33300 W/m²), as illustrated in Fig 5.4c. Since a severe local peak heat region is absent and the distribution is uniform, there will be an increase in

reliability and service life with the installation of a secondary reflector. The combined graphical representation of heat distribution in all the three orientations of the receiver tube in the system is shown in Fig 5.8

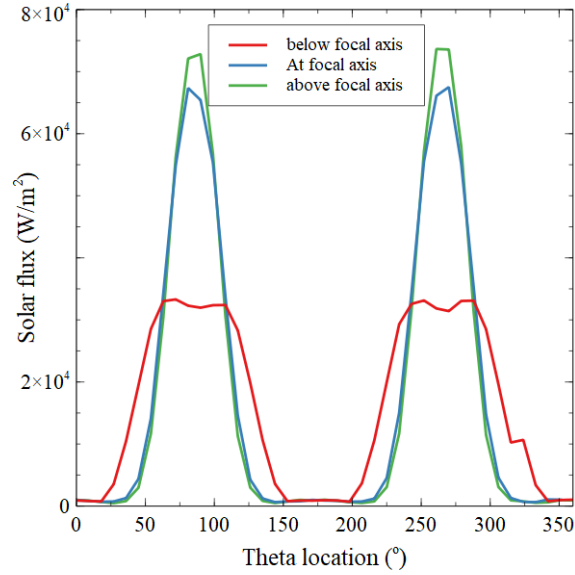


Fig 5.8: Heat flux distributions for the considered cases

As shown in Fig. 5.8, the heat flux gradient is significantly reduced with the secondary reflector and receiver tube placed below the focal axis of the primary collector. A dip in solar flux at the centre portion of the tube surface, as shown in Fig. 5.8, is due to the shadow cast by the receiver tube/secondary reflector. However, this region is smaller in the case when the receiver tube is placed below the focal axis of the primary reflector. This more uniform flux distribution is due to the secondary reflector's re-distribution of the solar radiation flux. It can enhance the reliability and longer service life of the PTC receiver.

CHAPTER 6: DESIGN OPTIMIZATION OF SECONDARY REFLECTOR

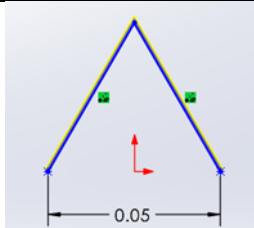
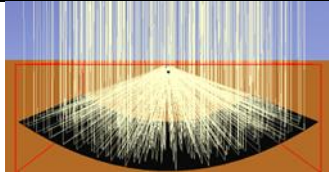
6.1 INTRODUCTION

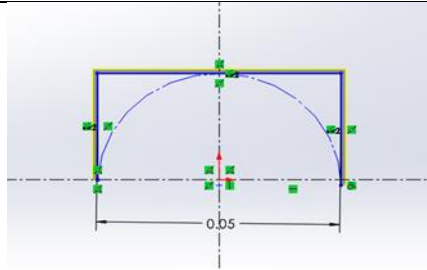
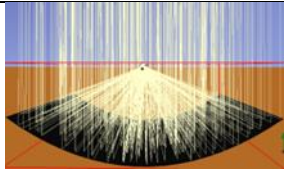
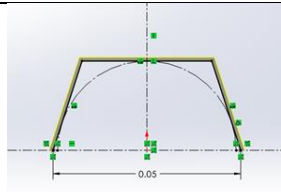
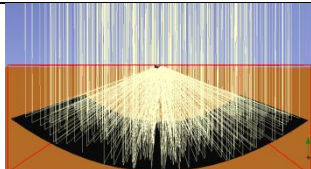
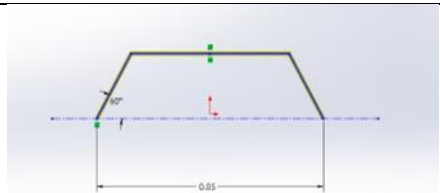
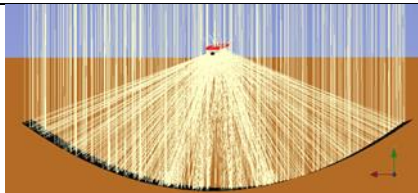
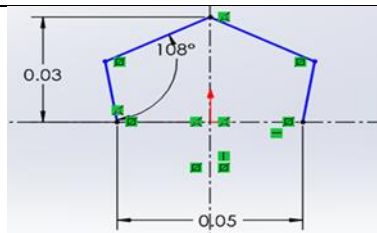
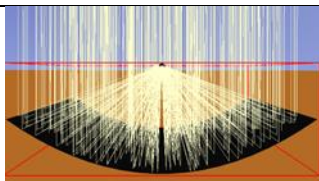
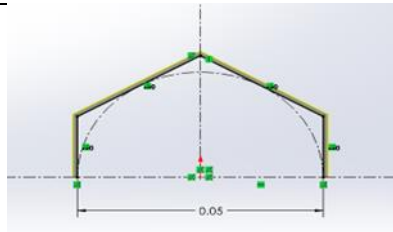
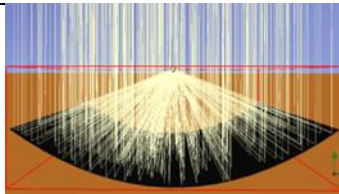
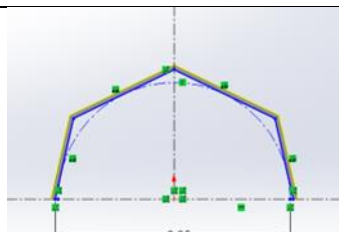
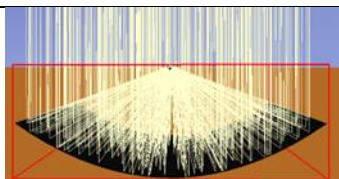
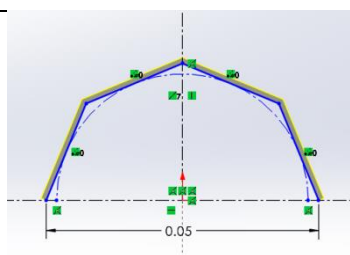
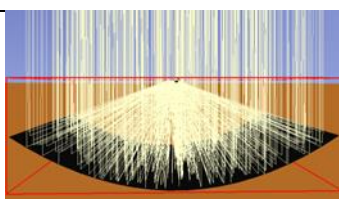
Performance of collectors may largely depend on the secondary-reflector profile design when small-aperture absorbers are used. Optimization of the secondary-reflector profile is an extremely challenging task because there is no established theory to ensure superior performance of derived profiles. In this work, an innovative optimization method is proposed to optimize the secondary-reflector profile .

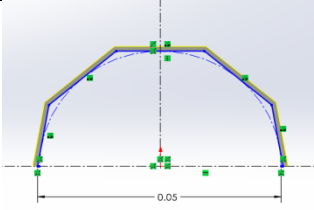
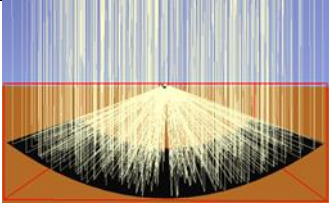
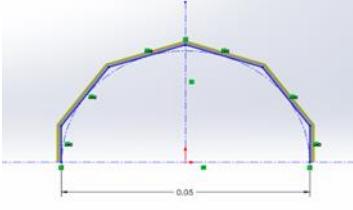
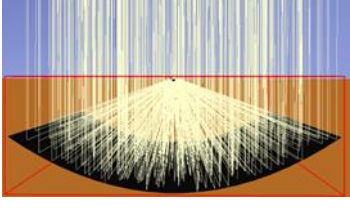
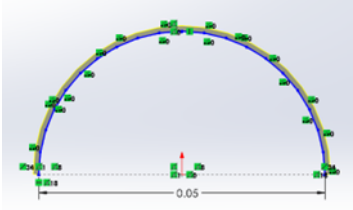
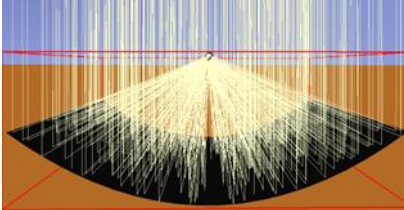
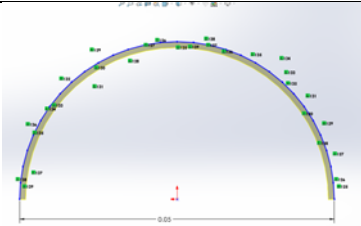
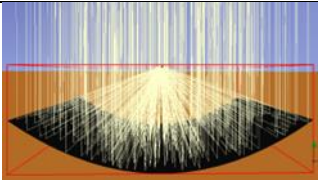
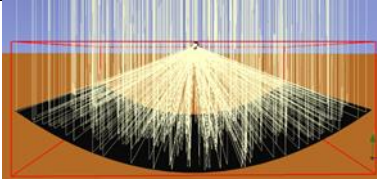
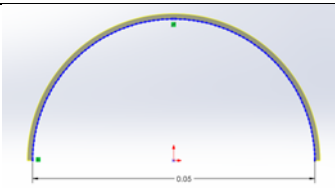
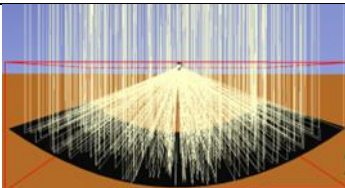
6.2 SHAPE SELECTION

The simulated configuration extended from 2 strips to an infinite strip (parabola), as shown in Table 6.1. The secondary reflectors' width is limited by twice the diameter of the secondary reflector, as a large secondary reflector width is superfluous and results in significant optical loss due to the shading effect. Thus, the aperture width of all secondary reflectors is fixed as 0.05 m. As the number of strips increases, the uniformity and maximum power output also increases. But in the case of the parabola, this parameter seems to be declining, so there is a point between where we will get the optimum values. Up to a point between 50 and 100 strips, the values increase, and beyond that, it declines.

Table 6.1 Optical simulations for different secondary reflector geometries

STRIP NAME	MODELS OF SECONDARY REFLECTOR	TONATIUH RAY TRACING
2 strip (60)		

3 strip (90^0)		
3 strip (108^0)		
3 strip (120^0)		
4 strip (108^0)		
4 strip (120^0)		
4 strip (128.57^0)		
4 strip (135^0)		

5 strip (140^0)		
6 strip (144^0)		
20 strip (171^0)		
30 strip (174^0)		
50 strip (176.4^0)		
100 strip (178.2^0)		

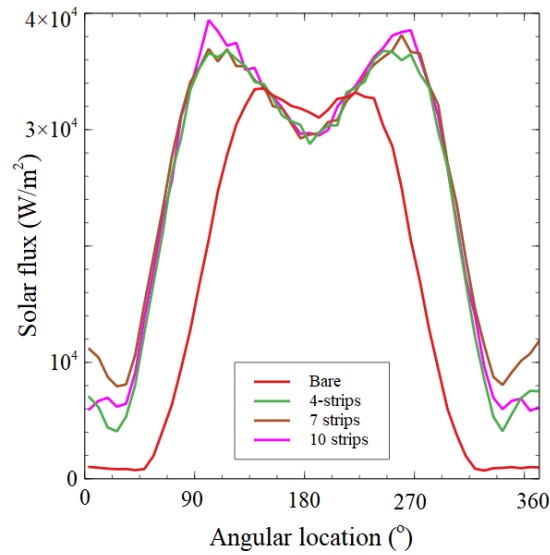
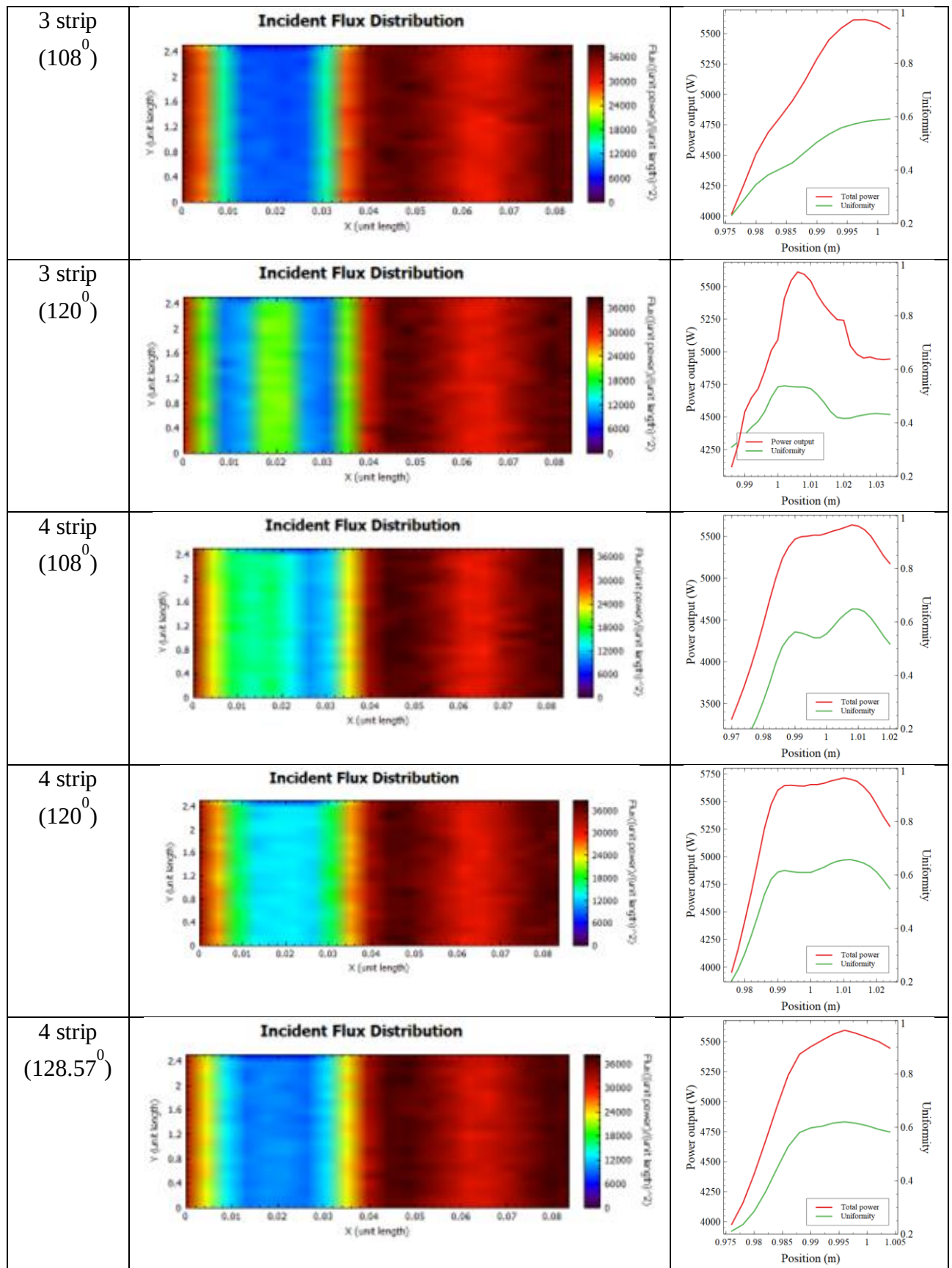


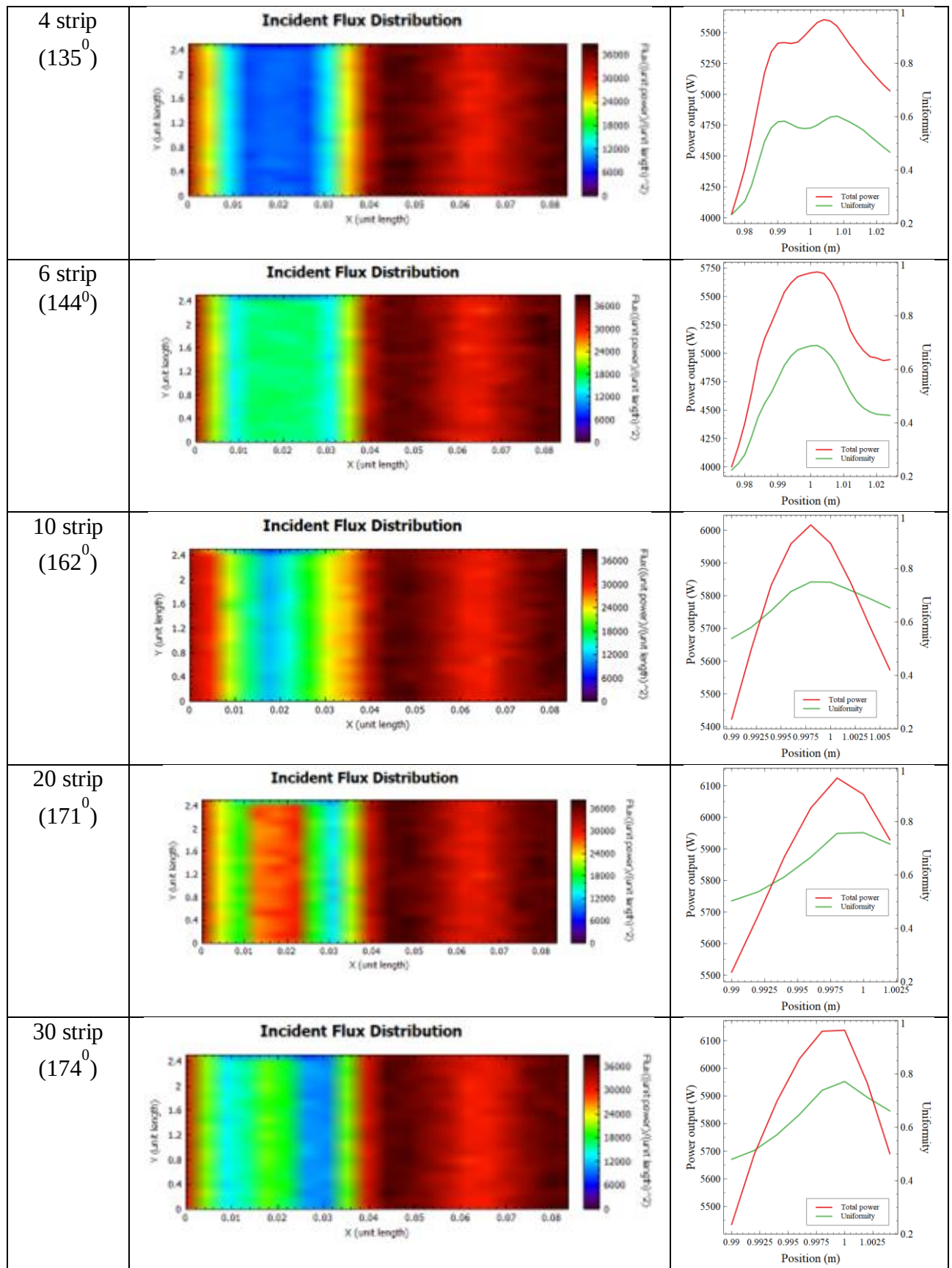
Fig 6.1: Flux distribution at periphery of receiver tube

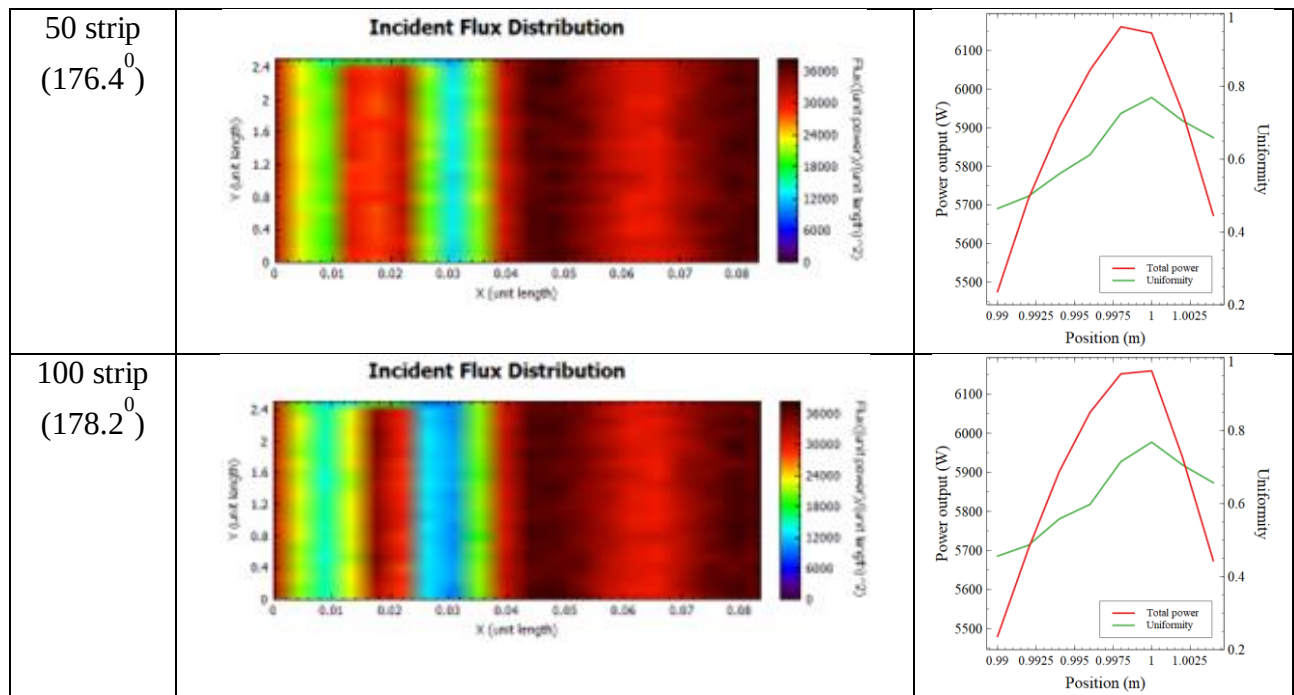
The solar flux distribution on the periphery of the receiver tube of a PTC without a secondary reflector and PTCs with three representative secondary reflectors is shown in Fig. 6.1. As shown in Fig. 6.1 the heat flux distribution of the PTC with secondary reflectors is significantly better as compared to that of the conventional PTC.

Table 6.2 Flux distribution, uniformity, and power output for different configurations

STRIP NAME	FLUX DISTRIBUTION	UNIFORMITY AND TOTAL POWER
2 strip (60°)	Incident Flux Distribution	
3 strip (90°)	Incident Flux Distribution	







The optical simulation analysis results for the different PTC configurations with the secondary reflector are described in Table 6.2. The secondary reflector position has been changed from 0.97m to 1.14m from the apex of the primary concentrator, and the corresponding results are illustrated in Table 6.2. The receiver tube position is fixed in such a way that the focal axis of the secondary reflector should lie on the axis of the receiver tube. The variation in power output and uniformity of heat flux distribution, as depicted in the graphs provided in Table 6.2 reveals that the secondary reflector geometry and position significantly influence the performance of the PTC. As the secondary reflector moves slightly away from the focal axis of the primary collector, the power output and uniformity are enhanced. Power output and uniformity showed the same trend in all cases and found that greater power output was obtained with better uniformity. However, the power output and uniformity decline when the secondary reflector moves further from the focal axis.

Among the different reflector shapes evaluated, the maximum uniformity achieved by the PTC with 4 strip 128.57° is 0.62, which is achieved with a secondary reflector position of 0.996 m from the apex of the primary concentrator. On the other hand, for the PTC with a 4 strip 120° secondary reflector, the maximum uniformity of 0.657 is achieved when the secondary reflector position is at 1.012m.

Table 6.3 Uniformity and output power for different secondary reflector configurations

Geometry	Maximum uniformity	Average Uniformity	Maximum Power (kW)
2 strip 60	0.602597	0.458810739	5.484
3 strip 90	0.626658	0.486523	5.712
3 strip 108	0.592036	0.462676786	5.614
3 strip 120	0.538562	0.447722328	5.257
4 strip 108	0.64884	0.470740212	5.637
4 strip 120	0.657695	0.55572708	5.714
4 strip 128.57	0.620359	0.4985206	5.595
4 strip 135	0.602293	0.508259739	5.605
6 strip 144	0.688938	0.609365444	5.716
10 strip 162	0.719162	0.56011364	6.016
20 strip 171	0.716103	0.616003111	6.124
30 strip 174	0.758756	0.60865925	6.137
50 strip 176.4	0.769579	0.62425225	6.161
100 strip 178.2	0.768378	0.61803225	6.159

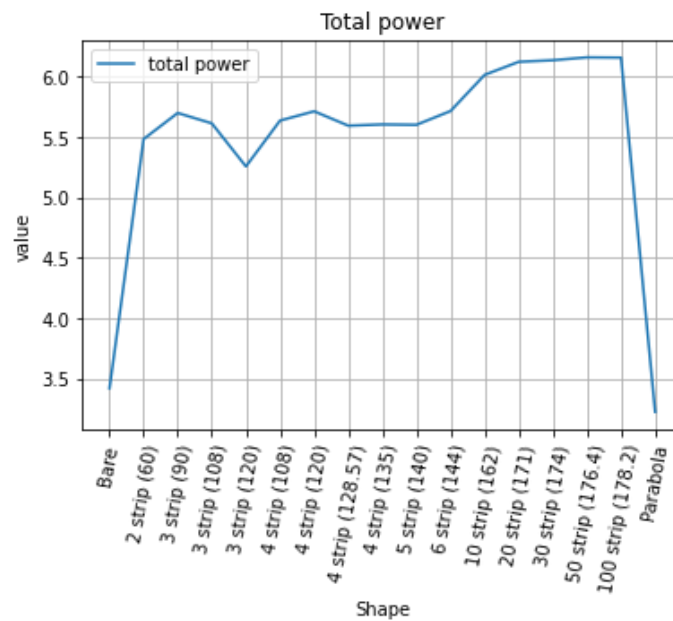


Fig 6.2 Power output with different shapes

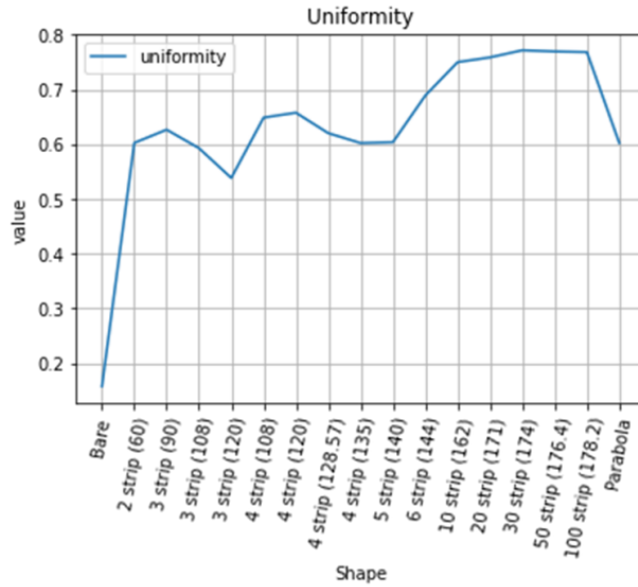


Fig 6.3a Uniformity with different shapes

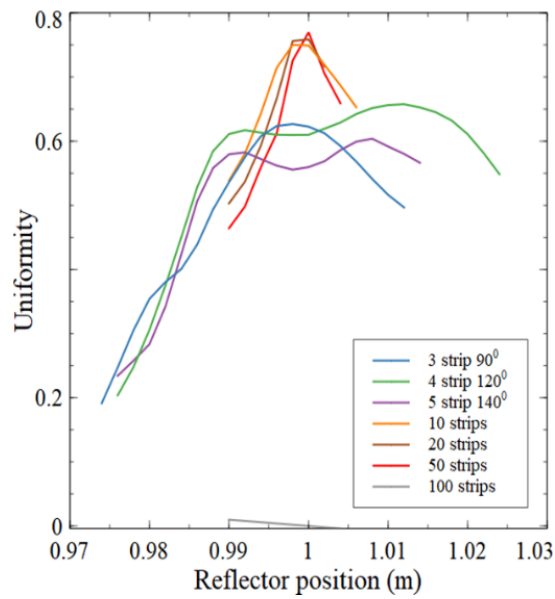


Fig 6.3b Uniformity at different reflector positions

It is observed that the maximum uniformity achieved by the PTC with a 50-strip secondary reflector is 0.769, which is the greatest among all other secondary reflectors. The maximum power output for the PTC is 6.161 kW. In general, the uniformity and power output increase as the number of strips of the secondary reflector increases.

CHAPTER 7: EXPERIMENTATION

7.1 WEATHER STATION

During experimental analysis the weather data has been collected from weather station at TKMCE. It measures global radiation, atmospheric temperature, humidity, wind speed, etc

Table 7.1 Experimental data (date 23/03/2022) for the secondary reflector with 6 strip (144°)

Time(h)	I_d (W/m ²)	V(m/s)	T_{amb} (°C)	T_{in} (°C)	T_{out} (°C)	Heat gain
9:00	24.2781	2.5	30.7	30.7	30.7	0
10:00	103.81	1.3	31.3	33.52	34.225	174.65
11:00	226.96	0.8	33.5	40.441	42.1095	375.8
12:00	270.25	1.3	33.4	52.97	54.65	471.83
13:00	390.46	1.3	32.6	69.71	70.25	500.63
14:00	396.33	2.2	31.8	81.36	82.63	551.4
15:00	290.52	1.9	30.7	87.88	89.71	336.45
16:00	20.11	3	29.5	88.35	89.64	228.23



Fig 7.1 Conventional PTC at TKMCE



Fig 7.2a PTC with selected shape (6 strip 144°)



Fig 7.2b PTC with selected shape (6 strip 144°)

PTC with selected shape (6 strip 144°), Flow rate=7m/s

Tank Capacity=50litres

15° change in angle with every 1 hr

Tubes are made of silicon, temperature is measured by T type thermocouple

CHAPTER 8: MACHINE LEARNING IN PTC

8.1 NEED FOR MACHINE LEARNING

Machine learning is a method to predict different outcomes of a complex system which are otherwise difficult to predict. This method is useful for systems which have complex interconnections between different parameters. Main aim is to find patterns between the variables. For systems which have many variables it will be difficult to represent through an equation in such cases we use different techniques in machine learning to find the patterns between input and results like ANN, SVM, etc

8.2 ARTIFICIAL NEURAL NETWORKS

Neural networks learn (or are trained) by processing examples, each of which contains a known "input" and "result," forming probability-weighted associations between the two, which are stored within the data structure of the net itself. The training of a neural network from a given example is usually conducted by determining the difference between the processed output of the network (often a prediction) and a target output. This difference is the error. The network then adjusts its weighted associations according to a learning rule and using this error value. Successive adjustments will cause the neural network to produce output which is increasingly like the target output. After a sufficient number of these adjustments the training can be terminated based upon certain criteria. This is known as supervised learning.

8.3 PARAMETER SETTINGS

- `hidden_layer_sizes{10, 12, 14}` default 12
- `activation{'identity', 'logistic', 'tanh', 'relu'}`, default = 'relu'
Activation function for the hidden layer .
- `solver{'lbfgs', 'sgd', 'adam'}`, default = 'adam'
- `alpha`, default = 0.0001
- `learning_rate{'constant', 'invscaling', 'adaptive'}`, default = 'constant'
- `learning_rate_init`, default = 0.001

- max_iter, default = 300
- shuffle, default = True

8.4 FEATURE STUDY

8.4.1 Heat Map of The Correlations

A heat map is a two-dimensional representation of information with the help of colours. It is used to visualize complex information here red colour indicates positive correlation and blue colour indicates negative correlation

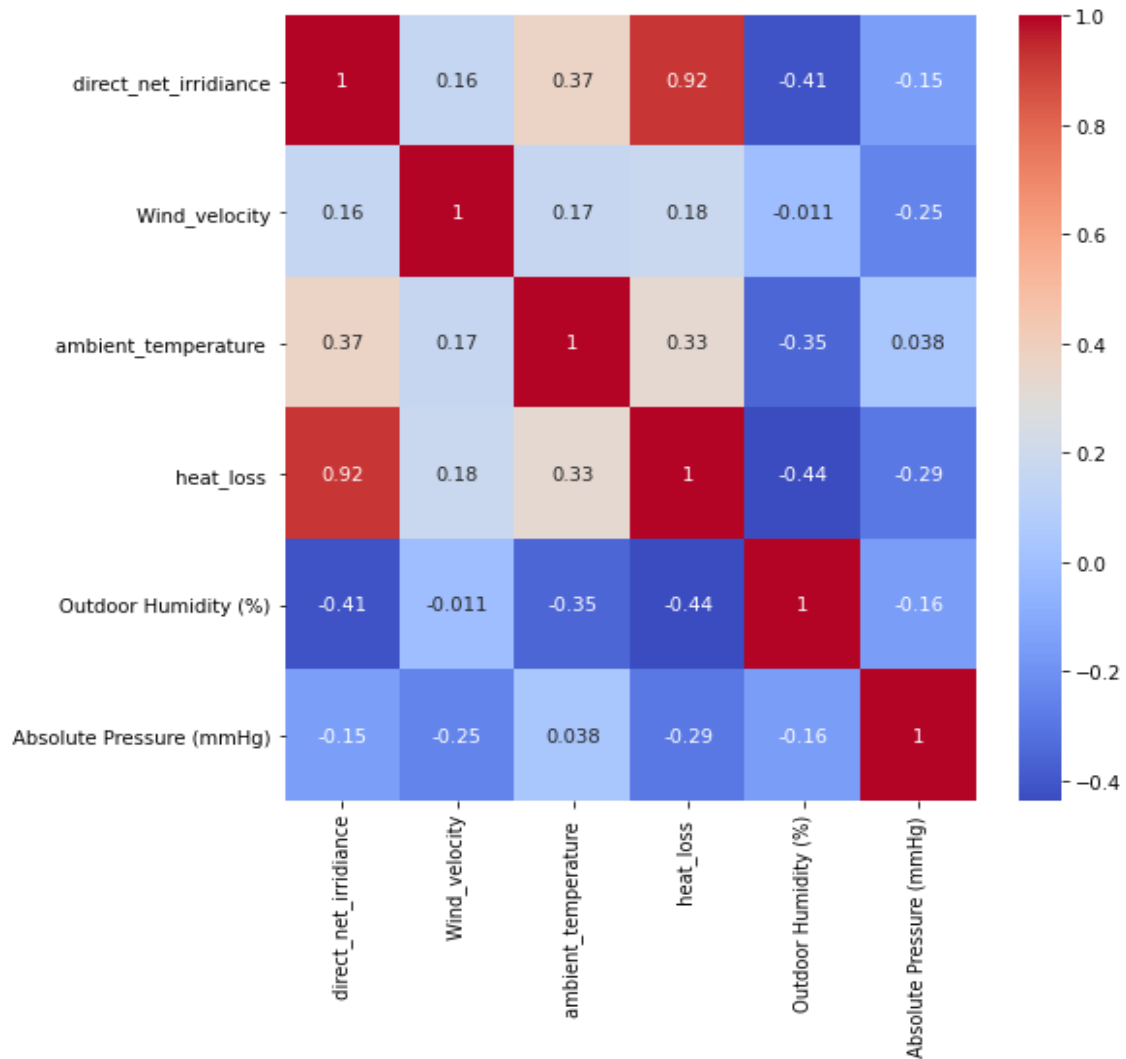


Fig 8.1: Heat map of the parameters taken for the ANN of the PTC

It's observed that the parameters of irradiance, heat loss and ambient temperature shows some correlation between each other. Other than that, there is a little correlation between other parameters.

8.4.2 Pairplot

Pair plot is used to understand the best set of features to explain a relationship between two variables or to form the most separated clusters. It also helps to form some simple classification models by drawing some simple lines or make linear separation in our dataset.

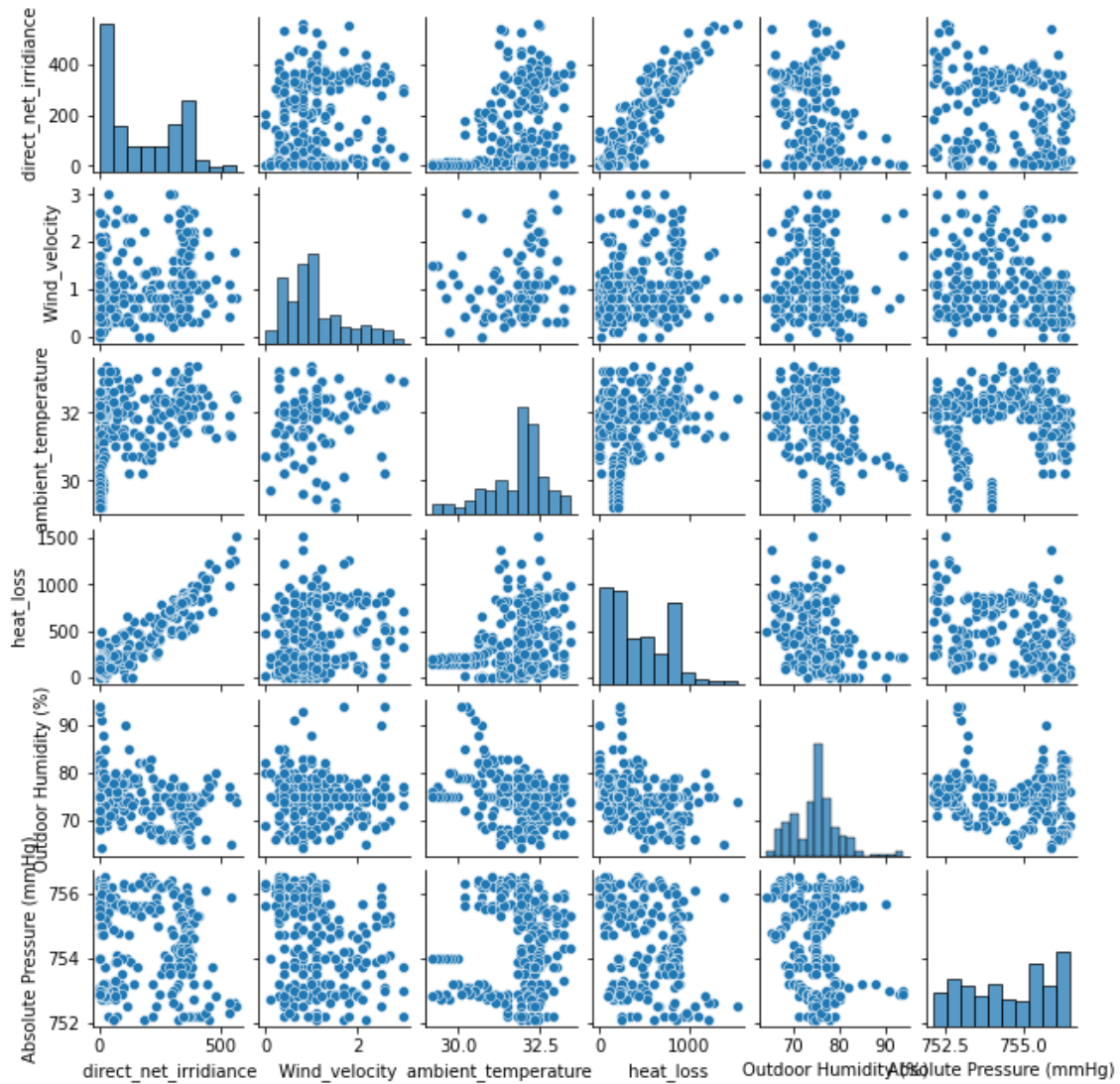


Fig 8.2: Pairplot of the parameters taken for the ANN of the PTC

- Linear separation is visible between irradiance and heat loss, parameters irradiance and heat loss can be replaced for prediction modelling

8.5 RESULTS AND OBSERVATIONS

After the analyse using ANN we get an accuracy of 60% for the model that we have formed using collected data of 5 days with 400 data points.

- It has been observed that sun's irradiance is the main contributing factor for the heat gain in a PTC, but due to its unpredictability mainly due to sudden clouding and raining there is huge fluctuations in heat gain that will get an accuracy for ANN and other machine learning algorithms not as expected as we wish
- Faced problems in finding weather patterns due to climate change which also caused a major problem in model predictions, due to these variations more detailed study in weather pattern is needed
- Next major challenge we faced is the collection of data, due to large variations in weather, data need to be collected for a large range of time, which need to cover all major weather conditions and patterns
- Correlations between the input parameters are complex and difficult to understand

8.6 SOLUTIONS

- Time intervals can be widened and take an average value which will helps in reducing the huge fluctuations in input parameters
- More varying seasons need to be taken for incorporating all the possible inputs
- Data generation techniques can be used to fill the missing gaps
- Input parameters can be collaborated and get more meaningful data sets
- More than one machine learning technique is to be used and need to be compared to get higher accuracy
- Large number of data is always helpful in achieving higher accuracy

- Extensive study in local weather patterns need to be conducted to get the clouding and raining pattern

8.7 FUTURE SCOPE

Studying about the patterns in nature is always a difficult task to apply mainly because of unpredictable nature of natural phenomena and complexity in studying natural factors due its complex interlinkages. So, because of it there is a need to study changing weather patterns for the past years and collaborate the study findings with the current prediction model.

Also, there is a scope in doing the prediction model with different ML algorithms and comparing it with each other.

CHAPTER 9: COST ANALYSES

Table 9.1 Cost analyses of the project

SL No	Type of expenditure	Rating/specification	Estimated Cost (Rupees)	Actual Cost (Rupees)
01	Fabrication of secondary reflector	Aperture width=0.05m Length=2.5m Material=Steel Sheet width=0.081m	1000	850
02	Printing	Poster Project report	500	750
03	Transportation	Petrol	200	150
		TOTAL	1700.00	1750.00

Before starting the project, cost estimation is done based on the market rate and demand for materials, products, and services. Estimated that a total of 1700.00 rupees are brought to purchase goods and services for the project. After completion of the project, the actual amount of expenditure is 1750.00 rupees. This is larger than the estimated cost. The actual expenditure is greater than the estimated price, it is because of those fluctuations in price, demand, taxes, transportation, printing errors, etc

CHAPTER 10: CONCLUSION

This chapter discusses the important conclusions drawn from the present study and discusses the various scopes for the future works.

9.1 CONCLUSION

Designed secondary reflector with different shapes in Solidworks, integrated with solar thermal collector system and optical performance analysis of parabolic trough solar collector has been conducted using Tonatiuh software. The power output of the PTC and uniformity of the distribution of solar flux over the absorber surface for different configurations has also been examined in this study. Also done a prediction modelling using ANN machine learning algorithm.

The salient conclusions from the above study are as follows:

- Heat flux distribution of PTC with secondary reflector is found to be more uniform than conventional PTC
- Regarding the configuration of the receiver tube, it is found that best and efficient result can be achieved when the receiver tube is placed below the focal axis with a secondary reflector installed on the system.
- The power output of the PTC, average heat flux and uniformity of heat flux distribution is better in the orientation in which the receiver tube is placed below the focal axis with a secondary reflector installed in the system.
- Power output and uniformity of PTC increases with increase in number of strips to a point between 50 and 100 strips after that it starts decreasing
- For avoiding sophisticated manufacturing techniques, we will select a shape which will have total power output of more than 5.5kW with significant difference in uniformity when compared to the conventional PTC
- For 6 strip (144°) shaped secondary reflector the highest obtained uniformity is 0.688938 at position 1.002m from the vertex of the collector at that point total power is 5.716kW
- Increase in uniformity with respect to conventional PTC is 77.066%
- Increase in total power output with respect to conventional PTC is 43.59%

- Conducted ANN and found an accuracy of 60%, found out that further detailed study is needed in areas of weather patterns and correlation of input parameters

From the results, it can be concluded that the heat flux distribution can be homogenised by moving the receiver toward the primary collector and adding a secondary reflector with the most desirable geometry and configurations. Also understands the need for a deeper study need to be conducted to do a prediction modelling of the system.

9.2 SCOPE FOR FUTURE WORK

There is a lot of scope in the field of solar thermal collector systems. In future, various studies can be conducted to increase the efficiency. They are as follows:

1. By varying the dimensions (length and diameter) of the absorber tube and glass envelope.
2. Using automatic tracking systems
3. The space between absorber tube and glass envelope can be evacuated to increase the efficiency
4. More detailed study in the prediction of heat gain and modelling using different machine learning techniques

9.3 SUMMARY

The depletion and environmental pollution caused by the excessive use of fossil fuels has led to the need to search for alternative energy sources. Solar energy is a renewable source that is clean and non-exhaustive in nature. Parabolic trough solar collector is the key technology to harness this energy. This project aims to design and develop a PTC with secondary reflector that gives the maximum power output keeping uniformity as the constrain. With an advanced optical simulation using Tonatiuh software, an effective shape for secondary reflector which can increase uniformity and power output is found and done machine learning to predict the heat gain for the system.

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